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CHAPTER 14

COMMUNICATION

SUGGESTED TIME: 5 PERIODS

IN THIS CHAPTER

This chapter explores the physics of modern communication systems: how information is represented in analogue and digital form, how radio waves propagate, how aerials and tuning circuits select signals, and how modulation, digitisation and communication channels (including satellites) make broadcasting and telephony possible.

- By the end of this chapter I can distinguish analogue from digital methods of representing information and state the advantages and disadvantages of each.
- By the end of this chapter I can describe the three modes of radio wave propagation (ground, sky and space waves) and match each to its frequency range and typical use.
- By the end of this chapter I can explain the roles of transmitting and receiving aerials and relate aerial length to wavelength.
- By the end of this chapter I can explain how an LC tuning circuit selects a station and interpret its resonance curve.
- By the end of this chapter I can explain the principles of modulation and demodulation and compare AM and FM.
- By the end of this chapter I can describe the operation of analogue-to-digital and digital-to-analogue converters, including sampling and quantisation.

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COURSE complete, illustrated, step by step

METHODS and worked examples

EXERCISES exam style + solutions

SUMMARY SHEET the essentials to remember

CONTENTS

1. Representing Information and Radio Wave Propagation
2. Aerials and the Tuning Circuit
3. Modulation, Demodulation, AM and FM
4. Digital Conversion and Communication Channels
5. Integration activity
6. Methods and worked examples
7. Exercises
8. Solutions to the exercises
9. Summary sheet

CHAPTER OBJECTIVES

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- By the end of this chapter I can explain the principles of modulation and demodulation and compare AM and FM.
- By the end of this chapter I can describe the operation of analogue-to-digital and digital-to-analogue converters, including sampling and quantisation.
- By the end of this chapter I can define bandwidth and sidebands and calculate the bandwidth of an AM signal.
- By the end of this chapter I can describe communication channels, the use of satellites and the role of base stations in mobile telephony.

PREREQUISITES

- Electromagnetic waves: nature, speed $c = 3.0 \times 10^8$ m/s, relation $c = f\lambda$ (Lower Sixth)
- Free oscillations and resonance in an LC circuit; natural frequency $f_0 = 1/(2\pi\sqrt{(LC)})$

(Lower Sixth)

- Alternating currents: peak and r.m.s. values, frequency, period (Lower Sixth)
- Wave phenomena: reflection, refraction, diffraction and superposition (Lower Sixth)
- Binary numbers and basic logic levels (useful for digital signals)

1 REPRESENTING INFORMATION AND RADIO WAVE PROPAGATION

Every evening in Yaoundé, Douala or Bamenda, millions of Cameroonians tune in to CRTV, listen to radio stations like Magic FM, or receive calls routed through MTN and Orange base stations — yet the voice of a presenter in Douala can be heard clearly in Wum, hundreds of kilometres away. How does invisible information travel through air, bend round the curve of the Earth, bounce off the sky, or hop to a satellite 36 000 km above the equator? Why does your radio crackle during a storm while a digital TV decoder keeps a stable picture? This chapter uncovers the physics behind these everyday miracles: how information is **represented**, carried by **radio waves**, modulated, digitised and channelled across Cameroon and the world.

In this first section we answer two questions: how is information represented (analogue or digital), and how does a radio wave physically travel from a transmitting mast to your receiver (ground wave, sky wave, space wave).

1.1 The Nature of Information in Communication

What is information?

In physics, **information** is any message that can be measured, stored or transmitted: the sound of a voice, a photograph, a text message, a video, or numerical data from a weather station on Mount Cameroon. None of these can travel far in its raw form. Sound waves die out after a few hundred metres; light from a picture cannot pass through walls.

The solution used in all communication systems is a three-stage chain:

1. a **transducer** converts the message into an electrical signal (a microphone turns sound into a varying voltage);
2. the electrical signal is impressed on a **carrier** — usually a high-frequency electromagnetic wave — which can travel enormous distances;
3. the carrier propagates through a **transmission medium** (the atmosphere, a cable, an optical fibre, or free space to a satellite) to a receiver, which extracts the original message.

KEY POINTS **The communication chain**

Information → electrical signal → carrier wave → transmission medium → receiver → original information. The carrier does the travelling; the information only "rides" on it.

1.2 The Analogue Method of Representing Information

Speak into an old telephone microphone and watch the voltage on an oscilloscope: the voltage rises and falls continuously, copying the shape of the sound wave. When the sound pressure doubles, the voltage doubles. The electrical signal is an analogy of the original message — hence the name.

DEFINITION **Analogue signal**

An **analogue signal** is a signal in which a physical quantity (voltage, current, amplitude) varies continuously with time, taking any value within a range, so that its variation mimics the variation of the original information.

Everyday examples of the analogue method:

- the **vinyl record**: the groove wiggles continuously, copying the sound waveform;
- the **analogue telephone**: the microphone voltage varies continuously with the voice;
- **AM and FM radio broadcasting**: the amplitude or frequency of the carrier varies continuously with the audio signal.

1.3 The Digital Method of Representing Information

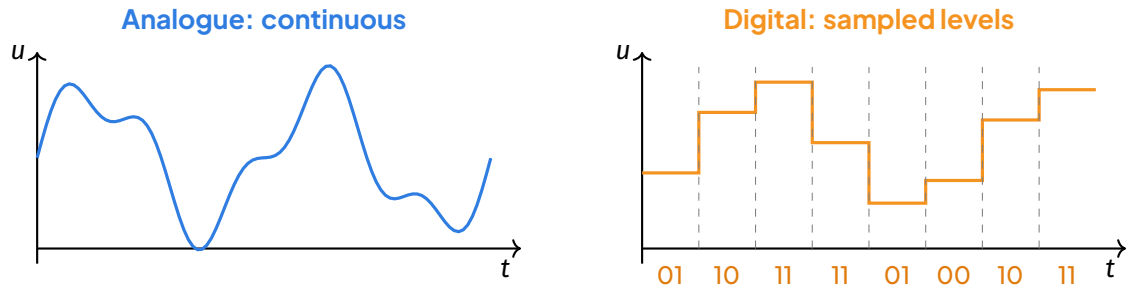
Now imagine instead that you measure the signal at regular instants and write down each measurement as a number, using only the digits 0 and 1. The smooth curve is replaced by a staircase of discrete levels, and each level is coded in **binary**.

DEFINITION **Digital signal**

A **digital signal** is a signal that can take only a finite number of discrete values, usually two (low = 0, high = 1). The original information is represented by a sequence of binary digits (**bits**).

Converting an analogue signal to digital requires two operations:

- **sampling**: measuring the value of the signal at regular time intervals (for a CD, 44 100 times per second);
- **quantisation**: rounding each measured value to the nearest allowed level, then coding that level in binary. With n bits per sample, 2^n different levels can be coded (for example 2 bits give $2^2 = 4$ levels: 00, 01, 10, 11).



An analogue waveform varies continuously; its digitised version is a staircase of quantised levels, each coded in binary.

1.4 Analogue versus Digital: Advantages and Disadvantages

Why did Cameroon switch television broadcasting from analogue to digital? The answer lies in how each method fights **noise** — the unwanted random disturbances (storm static, engine ignition, other stations) that add to any signal during transmission.

An analogue signal that picks up noise is damaged permanently: the receiver cannot tell which part of the wiggle is the music and which part is the crackle. A digital signal only needs the receiver to decide "0 or 1?". Even if noise distorts the pulse, as long as it is still recognisably a 1, the original value is recovered perfectly. Moreover, a digital **regenerator** can receive a weakened, noisy pulse train, decide each bit, and retransmit a brand-new, clean copy — something impossible for analogue amplifiers, which amplify the noise together with the signal.

Criterion	Analogue	Digital
Noise immunity	Poor: noise adds permanently to the signal	Excellent: bits can be regenerated perfectly
Copying / storage	Each copy degrades (tape hiss)	Perfect copies, easy storage on memory chips
Processing	Difficult	Easy: computers, error correction, encryption
Bandwidth required	Small	Large (many bits per second)
Circuit cost / complexity	Simple, cheap circuits	More complex electronics, needs converters

KEY POINTS **Key comparison**

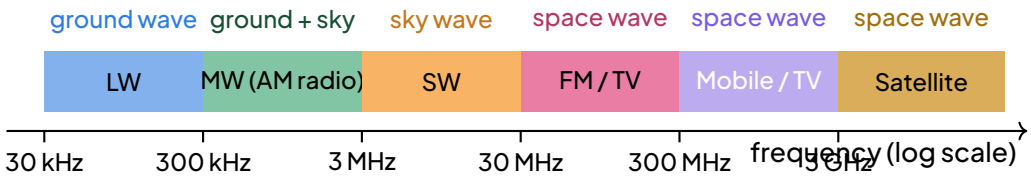
Digital wins on **noise immunity**, **regeneration**, **storage** and **processing**; analogue wins on **simplicity** and **low bandwidth**. This is why mobile phones, digital TV and the internet are digital, while a simple microphone lead remains analogue.

1.5 Radio Waves and the Electromagnetic Spectrum

Radio waves are electromagnetic waves — oscillating electric and magnetic fields travelling through space at the speed of light $c = 3.00 \times 10^8\ \mathrm{m\,s^{-1}}$. They occupy the low-frequency end of the electromagnetic spectrum, from a few kilohertz up to hundreds of gigahertz. Frequency and wavelength are linked by

$$c = f\lambda \qquad \Longleftrightarrow \qquad \lambda = \frac{c}{f}.$$

Different frequency bands behave very differently in the atmosphere, so each band is reserved for particular services:



The radio spectrum (logarithmic frequency scale) with typical uses and the dominant propagation mode of each band.

EXEMPLE **Wavelength of an FM station**

A Yaoundé FM station broadcasts at $f = 100$ MHz. Its wavelength is

$$\lambda = \frac{c}{f} = \frac{3.00 \times 10^8}{1.00 \times 10^8} = 3.0\ \mathrm{m}.$$

Compare this with a MW station at $f = 1.0$ MHz: $\lambda = 300$ m. Long waves bend around hills and buildings far better than short ones — a first clue to why the bands propagate differently.

1.6 Surface (Ground) Wave Propagation

A radio wave launched close to the Earth's surface does not travel only in a straight line: at low frequencies it **diffracts** (bends) around the curvature of the Earth and follows the ground, rather as water waves bend around a headland. This is the **surface wave** or **ground wave**.

DEFINITION **Surface (ground) wave**

The **surface wave** is a radio wave that travels along the Earth's surface, following its curvature by diffraction. It is effective for frequencies below about 3 MHz (long and medium waves) and can reach hundreds of kilometres over land and over a thousand kilometres over the sea, which is a good electrical conductor.

Diffraction is strong only when the wavelength is comparable to or larger than the obstacles and the scale of the curvature — hence the restriction to low frequencies. The ground wave is steadily absorbed by the soil, so its range is limited and it weakens over rough or poorly conducting terrain such as the rocky highlands of the North-West Region; over sea water, a much better conductor, the losses are smaller and the range is greater.

1.7 Sky Wave Propagation: Bending Back from the Ionosphere

High in the atmosphere, between roughly 60 km and 400 km above the ground, ultraviolet radiation from the Sun ionises the gases, creating a region of free electrons and ions called the **ionosphere**. Free electrons can be set oscillating by a passing radio wave and re-radiate it; the net effect is that the wave is progressively refracted — bent — back towards the ground.

PROPERTY **Reflection by the ionosphere**

The ionosphere bends back ("reflects") radio waves in the approximate range 3 MHz to 30 MHz (medium and short waves). Below this range the waves are largely absorbed; above it (FM, TV, mobile frequencies) the ionosphere is essentially transparent and the waves pass through into space. (The explanation of this frequency dependence is not examinable, but the property itself is.)

A sky wave returns to Earth at a considerable distance from the transmitter. The region between the limit of the ground wave and the point where the sky wave first lands is called the **skip distance** (or skip zone): a listener there receives nothing, while a listener much further away receives the station clearly. At night the lower, absorbing layers of the ionosphere weaken, so medium-wave stations can be heard over the whole country and even from abroad — which is why foreign stations appear on your MW dial after sunset.

WARNING **The ionosphere is not a mirror**

A very common mistake is to say the sky wave "bounces off the ionosphere like light off a mirror". In fact the ionosphere is a gradually varying medium: the wave is refracted — bent little by little as it penetrates deeper into the electron layer — until its path curves back down towards the Earth. There is no hard reflecting surface. In the exam, write "refraction (gradual bending) in the ionosphere", not "reflection off a solid layer".

1.8 Space Wave Propagation: Line of Sight

Above about 30 MHz the ionosphere lets the wave through and the wavelength is too short to diffract usefully around the Earth. The wave therefore travels essentially in a **straight line** from the transmitting aerial to the receiving aerial: this is the **space wave** (or direct wave), used by FM radio (88–108 MHz), television and mobile phone networks.

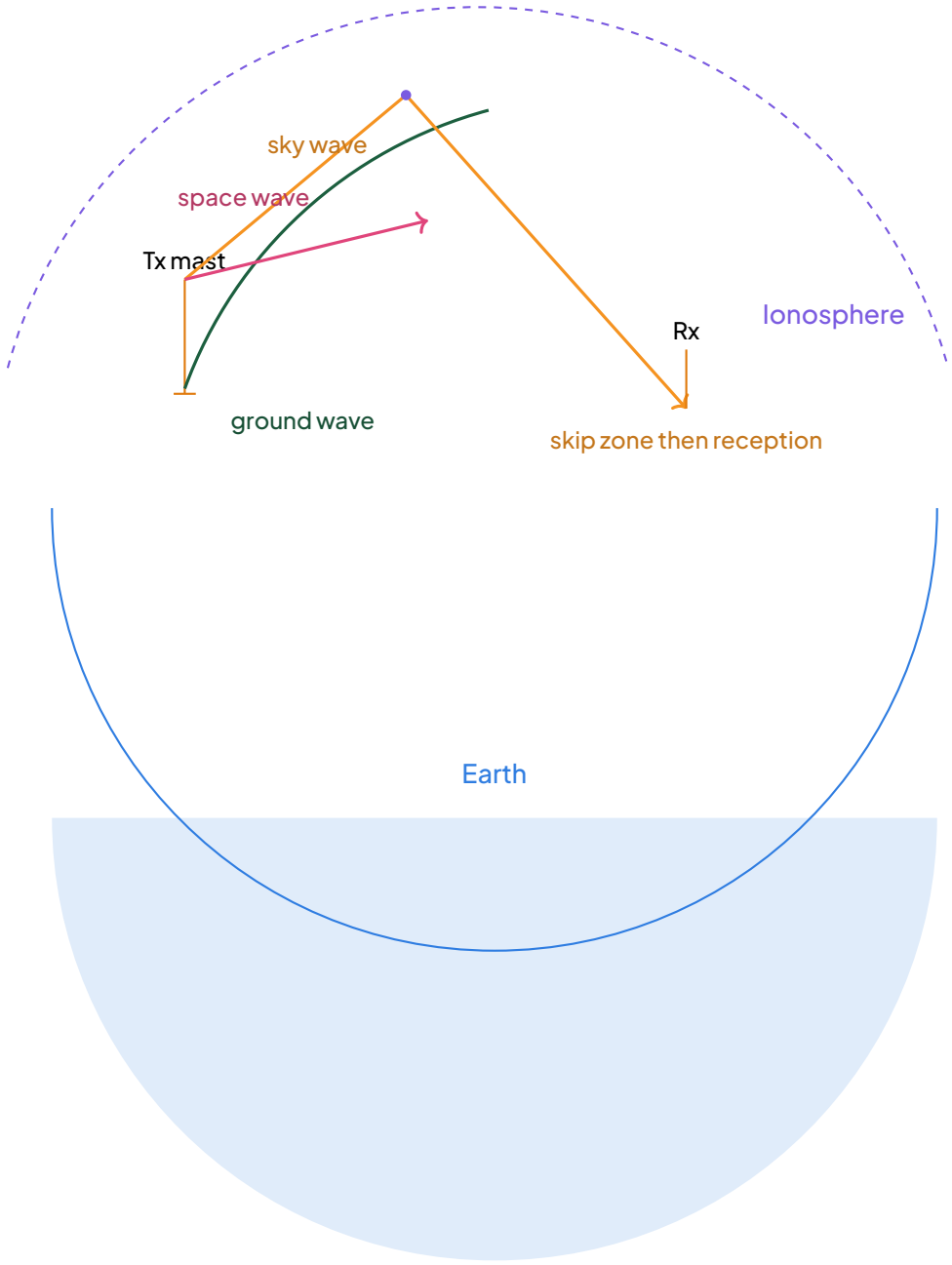
Because the Earth is curved, the straight-line path is blocked by the horizon. The maximum range is roughly the line-of-sight distance between the tops of the two aerials, which is why transmitters are placed on high masts and hills (for example on Mount Febe or Mount Cameroon)

and why networks of **relay masts** and base stations are needed to cover a country.

For a transmitting aerial of height h on a spherical Earth of radius R, the tangent from the top of the mast touches the Earth at the horizon. Pythagoras' theorem gives $(R + h)^2 = R^2 + d^2$, and since $h \ll R$,

$$d^2 = 2Rh + h^2 \approx 2Rh \qquad \Longrightarrow \qquad d \approx \sqrt{2Rh}.$$

With $R = 6.4 \times 10^6$ m and a mast of height $h = 100$ m, the horizon is at $d \approx \sqrt{2 \times 6.4 \times 10^6 \times 100} \approx 36$ km — the typical order of magnitude of an FM station's useful range.



The three propagation modes: the ground wave follows the Earth's curvature, the sky wave is refracted back by the ionosphere beyond the skip zone, and the space wave travels in a straight line limited by the horizon.

METHOD **Choosing the propagation mode**

Given a frequency f and a target distance d:

1. Compare f with the key limits 3 MHz and 30 MHz.
2. If $f < 3$ MHz: use the ground wave; reliable over a few hundred kilometres, day and night.
3. If $3\ \text{MHz} < f < 30\ \text{MHz}$: use the sky wave for long distances (hundreds to thousands of kilometres, especially at night); remember the skip zone — nearby receivers use the ground wave instead.
4. If $f > 30$ MHz: only the space wave works (line of sight). Check that d is less than the horizon distance; otherwise add relay masts or use a satellite link.

WORKED EXAMPLE 1 **Why FM fades but MW travels far**

Statement. A driver leaves Douala for Bafoussam listening to an FM station at 98 MHz. About 80 km from the city the signal fades away, yet a MW station at 1.2 MHz from the same city is still received clearly, and at night it can be heard in Garoua. Explain.

SOLUTION

Solution. The FM frequency, $f = 98\ \text{MHz} > 30\ \text{MHz}$, propagates only as a space wave: the ionosphere is transparent at this frequency and the wavelength ($\lambda = c/f \approx 3$ m) is far too short to diffract around the Earth. Once the car passes beyond the radio horizon of the transmitting mast — typically a few tens of kilometres — the straight-line path is blocked by the Earth's curvature and by hills, so reception is lost.
The MW frequency, $f = 1.2\ \text{MHz} < 3\ \text{MHz}$, travels as a ground wave that diffracts along the Earth's surface, giving coverage over hundreds of kilometres. In addition, at night the ionosphere refracts medium waves back to Earth (the absorbing lower layer disappears after sunset), so a sky wave can reach Garoua, more than 500 km away. Hence MW covers the whole country at night while FM cannot be received far beyond the horizon.

WARNING **Exam tip — GCE Advanced Level**

A favourite GCE question asks: "Explain why FM radio (≈ 100 MHz) cannot be received far beyond the horizon while MW radio can cover the whole country at night." Structure your answer in three points: (1) state the propagation mode available at each frequency; (2) for FM, say the ionosphere is transparent above 30 MHz and diffraction is negligible, so only line-of-sight space waves remain, limited by the horizon; (3) for MW, say the ground wave follows the Earth's curvature and the night-time ionosphere returns a sky wave to distant receivers. Quote $c = f\lambda$ if you compare wavelengths.

DID YOU KNOW? **Did you know?**

Short-wave sky waves can hop between the ionosphere and the ground several times, circling the globe. During the twentieth century, Cameroonians listened to international broadcasters such as the BBC on short wave thanks to these multi-hop paths. Today, the same physics limits your mobile phone (900 / 1800 MHz) to a few kilometres from the nearest base station — which is exactly why MTN and Orange must plant masts in every town and along the highways.

EXERCISE 1 **Propagation modes in Cameroon**

A station wants its programmes received (a) throughout the Littoral Region only, (b) throughout the whole country at night, (c) by a satellite uplink. For each case, choose a suitable frequency band and propagation mode, and justify your choice in two or three sentences.

SOLUTION **Exercise 1**

(a) An FM transmitter around 100 MHz on a high mast: the space wave gives clean, interference-free coverage up to the horizon, ideal for regional coverage. (b) A medium-wave transmitter around 1 MHz: the ground wave covers the region by day, and at night the ionosphere refracts a sky wave back to Earth over the whole country. (c) A microwave frequency of several GHz: far above 30 MHz, the wave passes straight through the ionosphere as a space wave and can reach a geostationary satellite.

2 AERIALS AND THE TUNING CIRCUIT

In the previous section we saw how information can be represented by analogue or digital signals, and how radio waves travel from place to place as ground waves, sky waves or space waves. But a radio wave does not appear from nowhere, and it does not deliver its message by itself. At the transmitting end, something must launch the electromagnetic wave into space; at the receiving end, something must capture it and something else must pick out one station among the hundreds that fill the air simultaneously over Yaoundé or Bamenda. These three tasks are performed by the **transmitting aerial**, the **receiving aerial** and the **tuning circuit**. They are the subject of this section.

2.1 The Transmitting Aerial

From current to wave.

Inside a transmitter, the information-bearing signal exists as a rapidly alternating current in a

wire. A current in a wire, however, keeps its fields close to the wire. To send energy across hundreds of kilometres, the current must be made to radiate: part of its electromagnetic field must detach itself and travel away as a self-propagating wave. This is precisely the job of the transmitting aerial.

DEFINITION **Aerial (antenna)**

An **aerial** (or **antenna**) is a conductor, or system of conductors, designed to convert the energy of a high-frequency alternating current into electromagnetic waves (transmitting aerial), or to convert the energy of an arriving electromagnetic wave back into an alternating e.m.f. (receiving aerial).

The physics is the following. A steady current produces only a static magnetic field, which does not radiate. An accelerating charge, however, produces changing electric and magnetic fields that regenerate each other and propagate away at speed $c = 3.00 \times 10^8\ \mathrm{m\,s^{-1}}$. In an aerial driven by a high-frequency alternating current, the electrons are continually accelerated back and forth, so radiation occurs continuously. The higher the frequency, the more violent the acceleration and the more efficient the radiation.

PROPERTY **Radiation efficiency and aerial size**

An aerial radiates efficiently only when its length is a significant fraction of the wavelength λ of the wave it is meant to launch. A wire much shorter than λ behaves almost like a lumped circuit element and radiates very poorly. (Proof not examinable; the result must be known and used.)

This explains a fact of everyday life. An audio signal at $f = 1\,\mathrm{kHz}$ has wavelength

$$\lambda = \frac{c}{f} = \frac{3.00 \times 10^8}{1.0 \times 10^3} = 3.0 \times 10^5\ \mathrm{m} = 300\ \mathrm{km},$$

so no practical aerial could radiate it directly. This is why we do not transmit the audio signal itself, but a high-frequency carrier (for example 1 MHz in the medium-wave band, $\lambda = 300$ m) that is modulated by the audio signal — the subject of the next section.

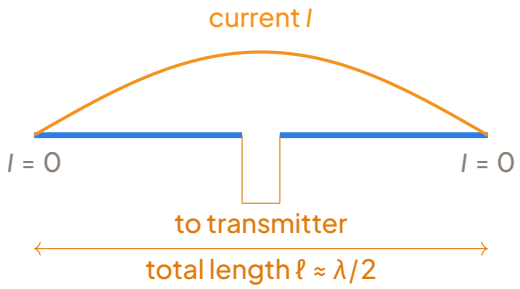
2.2 Aerial Dimensions and Wavelength

The half-wave dipole.

The simplest efficient aerial is the **half-wave dipole**: a straight conductor, split at its centre where the transmitter is connected, whose total length is approximately half a wavelength:

$$\ell \approx \frac{\lambda}{2} = \frac{c}{2f}.$$

At this length the aerial resonates: the current reflected from the open ends returns in phase, and a standing wave of current is established, with maximum current at the centre (where the feeder is connected) and zero current at the open ends. Resonance makes the dipole a strong radiator and gives it a convenient, purely resistive impedance at the feed point.



Half-wave dipole. The standing-wave current distribution (orange) is maximum at the centre feed point and zero at the open ends.

The quarter-wave monopole.

Where one half of the dipole is inconvenient, a **quarter-wave monopole** of length $\ell \approx \lambda/4$ is mounted vertically above a conducting ground plane (the Earth, or the metal body of a car). The ground plane acts as a mirror and supplies the "missing half" electrically. The telescopic rod of a portable radio and the whip aerial on a taxi in Douala are quarter-wave monopoles.

WARNING **Common mistake: λ instead of $\lambda/2$**

When a question asks for the length of a half-wave dipole at a given frequency, the answer is $\ell = \lambda/2 = c/(2f)$, not $\lambda = c/f$. Similarly a monopole is $\lambda/4$. Read the question twice and state which aerial you are using before computing.

WORKED EXAMPLE 2 **Length of a dipole for an FM station**

A student in Buea wants to build a simple half-wave dipole to receive an FM station broadcasting at $f = 98.5$ MHz. Calculate the required total length of the dipole, and the length of each arm.

SOLUTION

The wavelength is

$$\lambda = \frac{c}{f} = \frac{3.00 \times 10^{8}}{98.5 \times 10^{6}} = 3.05\ \text{m}.$$

For a half-wave dipole,

$$\ell = \frac{\lambda}{2} = \frac{3.05}{2} = 1.52\ \text{m}.$$

Each of the two arms therefore measures $\ell/2 \approx 0.76$ m. (In practice the length is shortened by about 5% to allow for end effects, giving ≈ 1.45 m, but the GCE expects $\lambda/2$.)

2.3 The Receiving Aerial

At the receiver, the process is reversed. The passing electromagnetic wave carries an oscillating electric field $\vec{E}$. This field exerts a force on the free electrons of the receiving conductor and drives them back and forth, inducing a small alternating **e.m.f.** at the same frequency as the wave.

PROPERTY **Receiving aerial**

A receiving aerial placed in an electromagnetic wave has an alternating e.m.f. induced in it, of the same frequency as the wave. The induced e.m.f. is greatest when the aerial is parallel to the electric field of the wave and has a resonant length (e.g. $\lambda/2$). The signal is very weak — typically microvolts to millivolts.

Here lies the central difficulty of radio reception: the air is filled with waves from many stations at once, plus television, mobile-phone and satellite signals. All of them induce e.m.f.s in the same aerial simultaneously. If the aerial were connected directly to an amplifier and loudspeaker, we would hear an unintelligible mixture of every station together. We need a device that selects one carrier frequency and rejects the others. That device is the tuning circuit.

2.4 The Tuning Circuit and Resonance

DEFINITION **Tuning circuit**

A **tuning circuit** is a parallel combination of an inductor L and a capacitor C connected to the receiving aerial. It selects one carrier frequency from all the signals induced in the aerial by means of electrical **resonance**.

DEFINITION **Resonance of the tuning circuit**

Resonance occurs when the frequency f of the incoming signal equals the natural frequency of the LC circuit,

$$f_0 = \frac{1}{2\pi\sqrt{LC}}.$$

At resonance the circuit responds to that frequency far more strongly than to any other.

Why the parallel LC selects a frequency.

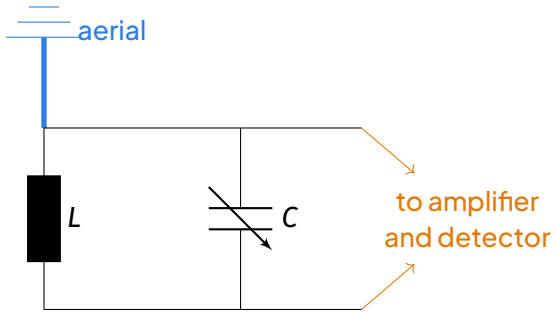
The inductor and capacitor exchange energy back and forth at the natural frequency f_0, just as a swing oscillates at its own natural frequency. Pushing a swing at exactly its natural rhythm builds up a large amplitude; pushing it at any other rhythm achieves little. In the same way, among all the e.m.f.s induced in the aerial, only the one whose frequency matches f_0 drives the LC circuit in step and builds up a large voltage across it.

More quantitatively, the inductor presents reactance $X_L = 2\pi f L$ and the capacitor $X_C = 1/(2\pi f C)$.

In parallel, at low frequency the inductor short-circuits the output (small X_L); at high frequency the capacitor short-circuits it (small X_C). Only near f_0, where $X_L = X_C$ and the two branch currents cancel, does the combination oppose the signal strongly.

PROPERTY **Impedance at resonance**

At resonance the impedance of a (ideal) parallel LC circuit is maximum — infinite for an ideal circuit, very large in practice. The weak current delivered by the aerial therefore produces its maximum voltage across the circuit precisely at f_0. This voltage is passed to the amplifier and demodulator. (The formula $f_0 = 1/(2\pi\sqrt{LC})$ must be known and used; its derivation is not examinable.)



Receiving aerial connected to a parallel LC tuning circuit with variable capacitor C. The voltage across the tank circuit is maximum at the resonant frequency.

Tuning to different stations.

Since $f_0 = 1/(2\pi\sqrt{LC})$, changing C changes the selected frequency. In a classical radio the "tuning knob" turns a variable capacitor: decreasing C increases f_0 (tunes to a higher-frequency station), increasing C decreases f_0.

METHOD **Finding the capacitance for a given station**

To tune to a station of carrier frequency f_0 with an inductor L:

1. Write the resonance condition $f_0 = \dfrac{1}{2\pi\sqrt{LC}}$.
2. Square both sides: $f_0^2 = \dfrac{1}{4\pi^2 LC}$.
3. Solve for the capacitance:
$$C = \frac{1}{4\pi^2 f_0^2 L}.$$
4. Substitute f_0 in hertz and L in henries; C comes out in farads (usually picofarads).

WORKED EXAMPLE 3 **Tuning a medium-wave receiver**

A tuning circuit uses an inductor of $L = 200\ \mu\text{H}$. Calculate the capacitance needed to receive a station broadcasting at $f_0 = 1.00\ \text{MHz}$.

SOLUTION

Using the method above:

$$C = \frac{1}{4\pi^2 f_0^2 L} = \frac{1}{4\pi^2\,(1.00 \times 10^6)^2\,(200 \times 10^{-6})}.$$

Computing the denominator: $4\pi^2 \approx 39.5$; $(1.00 \times 10^6)^2 = 1.00 \times 10^{12}$; so

$$C = \frac{1}{39.5 \times 1.00 \times 10^{12} \times 2.00 \times 10^{-4}} = \frac{1}{7.90 \times 10^9} \approx 1.27 \times 10^{-10}\ \text{F} = 127\ \text{pF}.$$

A variable capacitor covering roughly 127 pF at this setting will tune in the station.

WARNING **GCE exam tip: range of C for a whole band**

Numerical questions frequently give a fixed L and ask for the range of C needed to cover a whole band, e.g. the MW band 530 kHz to 1600 kHz. Because $C \propto 1/f_0^2$, the lowest frequency needs the largest capacitance and the highest frequency the smallest. Compute C_{max} at f_{min} and C_{min} at f_{max} — never the other way round.

EXERCISE 2 **Covering the FM band**

A tuning circuit has $L = 0.50\ \mu\text{H}$. Calculate the range of capacitance required to tune across the FM band from 88 MHz to 108 MHz.

SOLUTION **Exercise 1**

C_{max} corresponds to $f_{min} = 88\ \text{MHz}$:

$$C_{max} = \frac{1}{4\pi^2(88 \times 10^6)^2(0.50 \times 10^{-6})} = \frac{1}{39.5 \times 7.74 \times 10^{15} \times 5.0 \times 10^{-7}} \approx 6.5\ \text{pF}.$$

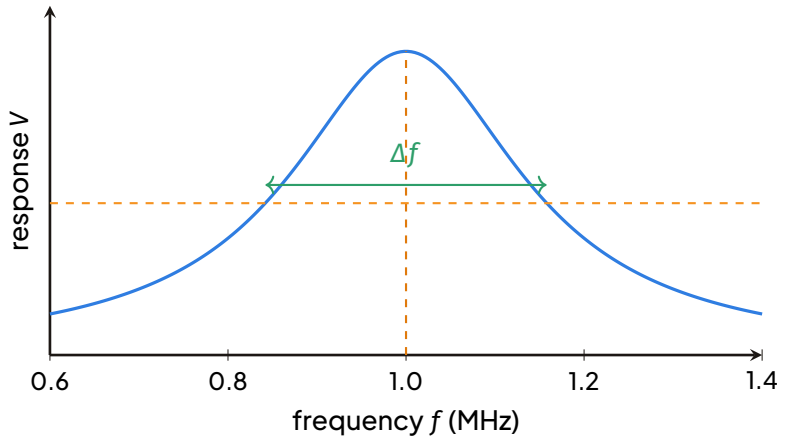
C_{min} corresponds to $f_{max} = 108\ \text{MHz}$:

$$C_{min} = \frac{1}{4\pi^2(108 \times 10^6)^2(0.50 \times 10^{-6})} \approx 4.3\ \text{pF}.$$

The variable capacitor must cover approximately 4.3 pF to 6.5 pF.

2.5 The Resonance Curve, Selectivity and Quality Factor

If we plot the voltage developed across the tuning circuit (its response) against the frequency of the incoming signal, we obtain the **resonance curve**: a sharp peak centred on f_0.



Resonance curve of a tuning circuit. The bandwidth Δf is measured between the frequencies where the response falls to $V_{max}/\sqrt{2}$ (half-power points).

DEFINITION **Bandwidth and quality factor**

The **bandwidth** Δf of the tuning circuit is the range of frequencies over which the response is at least $V_{max}/\sqrt{2}$ of its maximum (the half-power points). The **quality factor** is

$$Q = \frac{f_0}{\Delta f}.$$

A high Q means a narrow, sharp peak; a low Q means a broad, flat peak.

The selectivity–fidelity trade-off.

- **High Q (sharp resonance):** excellent selectivity — the circuit easily separates two stations with close carrier frequencies. But if the peak is too narrow, it clips the sidebands of the wanted station itself (we shall see in the next section that a modulated carrier occupies a band of frequencies, not a single one), and the sound loses its high notes: poor fidelity.
- **Low Q (broad resonance):** the whole sideband structure passes, giving good fidelity, but neighbouring stations may leak through: poor selectivity.

A practical receiver is designed with Q large enough to reject adjacent stations yet small enough to pass the full bandwidth of the wanted signal.

WORKED EXAMPLE 4 **Bandwidth of a tuning circuit**

A tuning circuit resonates at $f_0 = 1.00$ MHz with a quality factor $Q = 80$. Find its bandwidth and comment on its suitability for an AM station whose signal occupies ±5 kHz around the carrier.

SOLUTION

$$\Delta f = \frac{f_0}{Q} = \frac{1.00 \times 10^6}{80} = 1.25 \times 10^4 \text{ Hz} = 12.5 \text{ kHz}.$$

The AM signal needs a bandwidth of $2 \times 5 = 10$ kHz, which is less than 12.5 kHz. The circuit therefore passes the whole signal (acceptable fidelity) while still rejecting stations more than about 6 kHz away from the carrier. The design is adequate.

2.6 The Ferrite Rod Aerial (GCE favourite)

DID YOU KNOW? **Why portable MW radios use a ferrite rod**

Open a small portable medium-wave radio and you will not find a long wire: you will find a coil wound on a black rod of ferrite, a magnetic ceramic material. At 1 MHz the wavelength is 300 m, so a $\lambda/2$ dipole would be 150 m long — absurd for a pocket radio! Instead, the coil picks up the magnetic component of the wave. The ferrite rod, with its very high magnetic permeability, concentrates the magnetic flux of the passing wave through the coil, multiplying the induced e.m.f. by a large factor. The coil itself also serves as the inductor L of the tuning circuit, with a variable capacitor connected across it — aerial and tuner in one compact component. Because the rod responds to the magnetic field direction, reception is best when the rod is perpendicular to the direction of the transmitter, which is why rotating a portable radio changes the loudness.

KEY POINTS **Key points of this section**

- A transmitting aerial converts a high-frequency alternating current into an electromagnetic wave; it radiates efficiently only if its length is comparable to λ.
- Half-wave dipole: $\ell = \lambda/2$; quarter-wave monopole: $\ell = \lambda/4$; with $\lambda = c/f$.
- A receiving aerial has a small alternating e.m.f. induced in it by every passing wave; the tuning circuit selects one.
- A parallel LC circuit resonates at $f_0 = \dfrac{1}{2\pi\sqrt{LC}}$; at resonance its impedance is maximum, so the wanted station produces the largest voltage.

- Tuning is done with a variable capacitor: $C = \dfrac{1}{4\pi^2 f_0^2 L}$; largest C for the lowest frequency.
- The resonance curve peaks at f_0; bandwidth $\Delta f = f_0/Q$; high Q gives selectivity, low Q gives fidelity — a compromise is needed.
- Portable MW radios use a ferrite rod aerial to concentrate the magnetic flux into a small coil.

With one station now cleanly selected by the tuning circuit, the next task is to extract the actual information — music or speech — from the carrier wave. That is the work of demodulation, which we study in the next section together with the two great modulation schemes, AM and FM.

3 MODULATION, DEMODULATION, AM AND FM

In the previous section we saw how transmitting and receiving aerials launch and capture radio waves, and how the tuning circuit of a receiver selects one station by resonance. One question remains: how does the sound of a voice or of music actually travel on a radio wave? A pure radio wave of constant amplitude and constant frequency carries no information at all. The answer is **modulation**: we force a high-frequency radio wave to "carry" the low-frequency information signal by varying one of its properties. This section explains why modulation is unavoidable, describes the two classical methods — amplitude modulation (AM) and frequency modulation (FM) — and shows how the information is recovered at the receiver by demodulation.

3.1 Why Modulation is Necessary

Imagine a radio station in Douala trying to broadcast the audio signal from a microphone directly, without modulation. Three serious problems arise.

1. Audio frequencies radiate very poorly. The human voice and music occupy roughly the band 20 Hz to 20 kHz. Electromagnetic waves at such low frequencies are radiated extremely inefficiently by any practical aerial: the radiated power is proportional to $(l/\lambda)^2$ for a short dipole, so it falls off dramatically as frequency decreases (wavelength λ increases).

2. The aerial would be impossibly long. An efficient transmitting aerial has a length comparable to a quarter or a half of the wavelength, $\lambda = c/f$. For an audio frequency of $f = 1\,\text{kHz}$:

$$\lambda = \frac{c}{f} = \frac{3.00 \times 10^8}{1.0 \times 10^3} = 3.0 \times 10^5\ \text{m} = 300\ \text{km}.$$

A quarter-wave aerial would be 75 km long — longer than the distance from Yaoundé to Bafoussam! For a carrier at 100 MHz, $\lambda = 3$ m and a simple aerial of about 75 cm suffices.

3. All stations would overlap. If every station transmitted the same audio band (20 Hz–20 kHz), all broadcasts would occupy the same frequencies and no receiver could separate them. By giving each station its own high-frequency **carrier**, the tuning circuit of Section 2 can select one station and reject the others.

DEFINITION **Modulation, carrier wave, demodulation**

Modulation is the process of varying a property (amplitude, frequency or phase) of a high-frequency **carrier wave** in step with the low-frequency information (modulating) signal, so that the carrier "carries" the information. The result is the **modulated wave**. **Demodulation** (or detection) is the reverse process, carried out at the receiver, of extracting the information signal from the modulated carrier.

3.2 Amplitude Modulation (AM)

Intuition. Take a carrier $v_c = V_c \sin(2\pi f_c t)$ with f_c in the hundreds of kHz (medium wave) and make its amplitude rise and fall in step with the audio signal $v_m = V_m \sin(2\pi f_m t)$, where $f_m \ll f_c$. The "outline" (envelope) of the rapid oscillations then reproduces the shape of the audio signal.

PROPERTY **AM wave and modulation index**

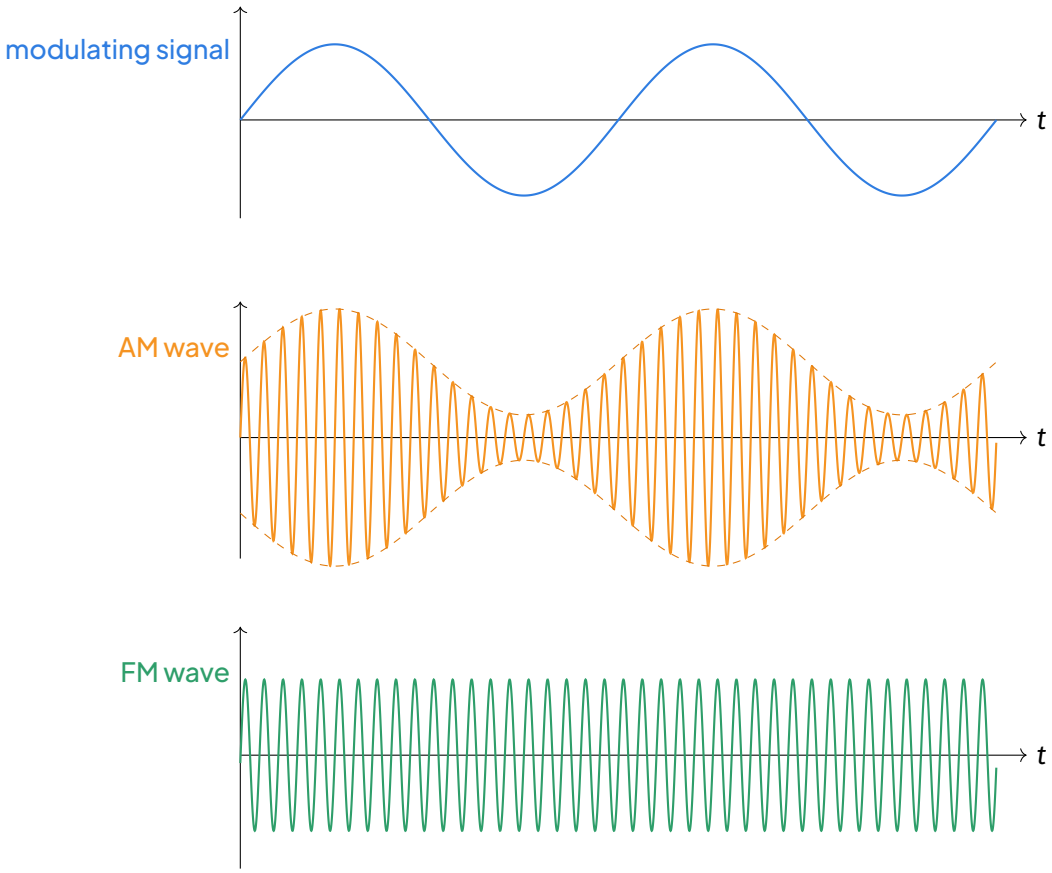
For a single-tone modulating signal, an amplitude-modulated wave has the form

$$v(t) = \left[V_c + V_m \sin(2\pi f_m t)\right] \sin(2\pi f_c t).$$

The **modulation index** (or depth of modulation) is

$$m = \frac{V_m}{V_c} = \frac{A_{\max} - A_{\min}}{A_{\max} + A_{\min}},$$

where $A_{\max}$ and $A_{\min}$ are the maximum and minimum peak-to-peak values of the envelope. For undistorted AM, $0 \le m \le 1$ (often quoted as a percentage, $m \times 100\,\%$). If $m > 1$ the envelope reaches zero and the signal is **overmodulated**: the envelope no longer matches the information signal and the received sound is distorted.



The same modulating (audio) signal impressed on a carrier: in AM the amplitude follows the signal (dashed envelope); in FM the frequency rises and falls while the amplitude stays constant.

PROPERTY **Frequency spectrum of an AM wave — sidebands**

Using the identity $\sin A \sin B = \frac{1}{2}[\cos(A - B) - \cos(A + B)]$, the AM wave expands to three frequency components:

$$v(t) = V_c \sin(2\pi f_c t) + \frac{mV_c}{2} \sin[2\pi(f_c + f_m)t] - \frac{mV_c}{2} \sin[2\pi(f_c - f_m)t].$$

The spectrum consists of:

- The **carrier** at f_c (amplitude V_c, unchanged).
- An **upper sideband** (USB) at $f_c + f_m$ (amplitude $mV_c/2$).
- A **lower sideband** (LSB) at $f_c - f_m$ (amplitude $mV_c/2$).

The **bandwidth** required is $B = 2f_m$. For audio up to 5 kHz, $B = 10$ kHz.

PROPERTY **Power in an AM wave**

The total transmitted power P_t (into a load R) is the sum of carrier and sideband powers:

$$P_t = P_c + P_{USB} + P_{LSB} = \frac{V_c^2}{2R} + \frac{(mV_c/2)^2}{2R} + \frac{(mV_c/2)^2}{2R} = P_c\left(1 + \frac{m^2}{2}\right).$$

At $m = 1$ (maximum undistorted modulation), the sidebands carry $1/3$ of the total power and the carrier $2/3$. The carrier itself conveys no information; this is a fundamental inefficiency of AM.

METHOD **Reading the modulation index from an oscilloscope trace**

1. Display the AM wave on an oscilloscope (time base slow enough to see the envelope clearly).
2. Measure A_{max}, the peak-to-peak height of the envelope at its widest, and A_{min}, the peak-to-peak height at its narrowest.
3. Compute $m = \dfrac{A_{max} - A_{min}}{A_{max} + A_{min}}$.
4. Check: if $A_{min} = 0$ then $m = 1$; if the trace pinches below zero (envelope crosses itself), the wave is overmodulated ($m > 1$).

WORKED EXAMPLE 5 **Modulation index of a broadcast signal**

An oscilloscope connected to an AM transmitter shows an envelope whose maximum height is $A_{max} = 9.0$ cm and minimum height $A_{min} = 3.0$ cm. (a) Find the modulation index. (b) The carrier amplitude is $V_c = 5.0$ V; find the amplitude V_m of the modulating signal.

SOLUTION

Solution. (a) Apply the definition:

$$m = \frac{A_{max} - A_{min}}{A_{max} + A_{min}} = \frac{9.0 - 3.0}{9.0 + 3.0} = \frac{6.0}{12.0} = 0.50.$$

The wave is 50 % modulated, comfortably below overmodulation.
(b) Since $m = V_m/V_c$,

$$V_m = mV_c = 0.50 \times 5.0 = 2.5\ \text{V}.$$

3.3 Frequency Modulation (FM)

In frequency modulation the amplitude of the carrier is kept strictly constant; it is the instantaneous frequency that varies in step with the information signal:

$$f(t) = f_c + k\,v_m(t),$$

where k is a constant of the modulator (Hz/V). When the audio signal is at its positive peak the frequency is highest; at the negative peak it is lowest. The maximum departure from the carrier frequency is called the **frequency deviation** Δf (for FM broadcast radio, $\Delta f = 75$ kHz around a carrier in the band 88 MHz–108 MHz). The loudness of the sound is encoded in the size of the frequency swing ($\Delta f \propto V_m$), and the pitch of the sound in the rate of the swing (f_m).

WARNING **Common mistake: do not vary the FM envelope**

A very frequent error in sketches is to draw the FM wave with a varying amplitude like AM. In FM the envelope is flat: every cycle has the same height. Only the spacing between cycles (the instantaneous frequency) changes. Examiners penalise FM waves drawn with a wavy envelope.

PROPERTY **Bandwidth of an FM wave — Carson's rule**

Unlike AM, an FM wave has an infinite number of sidebands (Bessel functions). However, most power is contained within a bandwidth approximated by **Carson's rule**:

$$B \approx 2(\Delta f + f_m).$$

For commercial FM radio ($\Delta f = 75$ kHz, $f_m \le 15$ kHz), $B \approx 180$ kHz. Channels are spaced 200 kHz apart. This wide bandwidth is the price paid for FM's noise immunity and high fidelity.

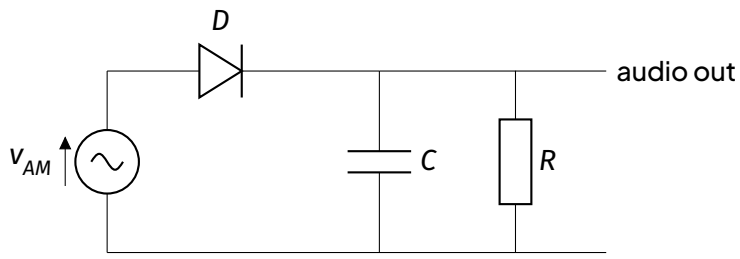
3.4 Demodulation: Recovering the Information

At the receiver, after the tuning circuit has selected the station and amplifiers have strengthened the signal, the information must be separated from the carrier.

AM: the envelope (diode) detector. The AM signal is passed through a diode, which removes the negative half-cycles (half-wave rectification). The remaining pulses are smoothed by a capacitor C in parallel with a resistor R (the load): the capacitor charges to each peak and discharges slowly through R between peaks, so the output voltage follows the envelope — that is, the original audio signal. The time constant must satisfy

$$\frac{1}{f_c} \ll RC \ll \frac{1}{f_m},$$

so that the carrier ripple is removed but the audio variations are preserved.

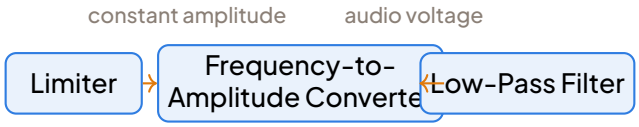


Simple diode envelope detector: the diode rectifies the AM wave; the *RC* network smooths it so the output follows the envelope (the audio signal).

FM: principle of detection. Since the information in FM lies in the frequency variations, an FM detector converts frequency changes into amplitude changes, and then recovers the audio signal. Classic methods include:

- **Slope detection**: a tuned circuit deliberately detuned so the signal sits on the slope of its resonance curve; frequency changes become amplitude changes.
- **Foster–Seeley discriminator** / **Ratio detector**: balanced circuits giving a voltage proportional to frequency deviation.
- **Phase-Locked Loop (PLL)**: modern integrated-circuit method where a VCO locks to the input frequency; the control voltage is the demodulated audio.

Crucially, before detection the FM wave passes through a **limiter** stage (a saturated amplifier) that clips the wave to a fixed amplitude, removing all amplitude variations caused by noise.



Block diagram of an FM demodulator: the limiter removes amplitude noise, the converter translates frequency swings into voltage swings, and the low-pass filter removes residual carrier components.

3.5 AM versus FM: Advantages and Disadvantages

PROPERTY **Why FM sounds cleaner**

Most natural and man-made interference — lightning flashes, sparks from car ignition systems, electric motors — adds unwanted amplitude variations to the received wave. An AM receiver interprets these as part of the signal, so they are heard as crackles and hiss. An FM receiver first passes the wave through a limiter that removes all amplitude variations, so this noise is rejected; only the frequency variations, which carry the programme,

survive. This is why FM radio sounds noticeably cleaner than AM.

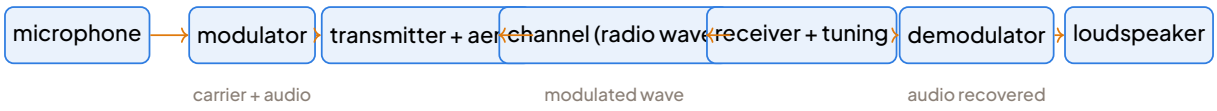
	AM	FM
Property varied	Carrier amplitude	Carrier frequency
Bandwidth	Narrow ($\approx 2f_m$, about 10 kHz)	Wide ($\approx 2(\Delta f + f_m)$, about 200 kHz)
Noise immunity	Poor; crackle and hiss from amplitude noise	Excellent (limiter rejects amplitude noise)
Sound quality	Lower fidelity (restricted audio bandwidth)	High fidelity (wide audio band, stereo capable)
Range	Long: ground and sky waves, beyond horizon	Shorter: space wave, roughly line of sight
Transmitter efficiency	Low (carrier wastes power)	Higher (constant amplitude, class C amps)
Cost/complexity	Simple, cheap transmitters and receivers	More complex, costlier equipment

WARNING **GCE exam tip**

"Compare AM and FM, giving one advantage of each over the other" is a frequent structured-question item at Advanced Level. Learn the comparison table above, and always justify: AM's long range comes from ground/sky-wave propagation at lower carrier frequencies (MF/HF); FM's quality comes from noise limiting and its large bandwidth, which is also why FM needs many more channels and is restricted to VHF line-of-sight coverage.

3.6 The Complete Radio Communication Chain

We can now assemble the whole system studied in this chapter into a single block diagram.



Block diagram of a complete radio communication system: the audio signal modulates a carrier, the modulated wave crosses the channel, and the receiver selects, amplifies and demodulates it to restore the sound.

KEY POINTS **Key points**

- Modulation is necessary because audio frequencies radiate poorly, would need kilometre-long aerials, and would make all stations overlap.
- AM: the carrier amplitude follows the signal; $m = \dfrac{A_{max} - A_{min}}{A_{max} + A_{min}}$, with $m \leq 1$ to avoid overmodulation. Spectrum: carrier ± two sidebands; bandwidth $2f_m$. Power $P_t = P_c(1 + m^2/2)$.
- FM: the carrier frequency follows the signal; amplitude is constant; maximum frequency swing is deviation Δf. Bandwidth $\approx 2(\Delta f + f_m)$ (Carson's rule).
- AM demodulation: diode envelope detector with RC smoothing ($1/f_c \ll RC \ll 1/f_m$).
- FM demodulation: limiter + frequency-to-amplitude converter (discriminator, ratio detector, or PLL).
- AM: cheap, long range (ground/sky wave), but noisy and low fidelity. FM: noise-immune, high fidelity, but wide bandwidth and line-of-sight range.

EXERCISE 3 **AM and FM reasoning**

1. An AM transmitter has a carrier of amplitude 8.0 V modulated by an audio signal of amplitude 6.0 V. Calculate the modulation index and state whether the transmission is distorted.
2. Explain, with reference to the limiter stage, why a thunderstorm causes crackles on an AM radio but barely affects an FM receiver.
3. An FM station broadcasts on a carrier of 101.5 MHz with a deviation of ±75 kHz. Between which frequencies does the instantaneous frequency vary?
4. (Challenge) An AM broadcast station operates at 1000 kHz with a maximum audio frequency of 5 kHz. What are the frequencies of the upper and lower sidebands? What is the bandwidth?

SOLUTION **Exercise 1**

1. $m = V_m/V_c = 6.0/8.0 = 0.75$. Since $m < 1$, the modulation is undistorted.
2. Lightning produces electromagnetic pulses that add spurious amplitude variations. The AM receiver's envelope detector reads these as signal, hence crackles. The FM

receiver's limiter clips all amplitude variations before detection; only frequency variations carry the programme, so the interference is rejected.

3. f varies between $f_c - \Delta f = 101.5 - 0.075 = 101.425$ MHz and $f_c + \Delta f = 101.575$ MHz.
4. USB: $1000 + 5 = 1005$ kHz. LSB: $1000 - 5 = 995$ kHz. Bandwidth $= 2 \times 5 = 10$ kHz.

DID YOU KNOW? **Radio in Cameroon**

Cameroon's CRTV radio and many private stations broadcast both on medium wave (AM) and on FM. In large towns like Douala and Yaoundé listeners prefer FM for its clear, stereo sound, but in remote rural areas of the North or East regions medium-wave AM is still valued because its ground and sky waves reach villages far beyond the horizon where no FM transmitter is visible. The next section will show how digital techniques and satellites extend this coverage even further.

4 DIGITAL CONVERSION AND COMMUNICATION CHANNELS

In the previous section we saw how a message is carried by a high-frequency carrier through **amplitude modulation** or **frequency modulation**. But before a signal even reaches the transmitter, modern communication systems usually convert it into **digital** form — a stream of numbers. Your mobile phone in Yaoundé does not send your voice as a continuously varying wave; it sends numbers, coded in binary, at a rate of thousands of numbers per second. In this final section we study how analogue signals are converted to digital and back, why engineers prefer digital transmission, and how channels — cables, radio links, optical fibres and satellites — carry the information to its destination.

4.1 Analogue and Digital Signals

DEFINITION **Analogue and digital signals**

An **analogue signal** varies continuously in time and can take any value within a range (example: the voltage from a microphone).
A **digital signal** can take only a finite number of discrete values, usually two ("0" and "1", i.e. low/high voltage). The information is carried as a sequence of **bits** (binary digits).

An analogue signal carries its information in the exact shape of the wave; any noise added to that shape is indistinguishable from the message. A digital signal carries its information in a code; as long as a receiver can still tell a "0" from a "1", the message is perfectly recovered.

This single difference explains almost all the advantages of digital transmission studied below.

4.2 Analogue-to-Digital Conversion (ADC)

The problem. An analogue signal varies continuously in time and can take any value within a range. A computer or a digital transmitter can only handle numbers with a finite number of digits. Conversion therefore requires three steps: **sampling**, **quantisation** and **coding**.

Step 1: Sampling. Instead of describing the signal at every instant, we measure ("sample") its value at regular time intervals T_s, called the **sampling period**.

DEFINITION **Sampling frequency**

The **sampling frequency** f_s is the number of samples taken per second:

$$f_s = \frac{1}{T_s} \qquad \text{(unit: hertz, Hz).}$$

How fast must we sample? If we sample too slowly, rapid wiggles of the signal are missed and the reconstructed signal is distorted (an effect called aliasing). The answer is given by the Nyquist criterion.

PROPERTY **Nyquist (sampling) criterion**

To reconstruct a signal faithfully, the sampling frequency must be at least twice the highest frequency present in the signal:

$$\boxed{f_s \geq 2\,f_{\max}}$$

The minimum value $f_s = 2f_{\max}$ is called the Nyquist rate. (Proof not examinable; the idea is that at least two points per cycle of the fastest component are needed to capture it.)

EXEMPLE **Choosing a sampling frequency**

The human ear perceives frequencies up to about $f_{\max} = 20\,\text{kHz}$. To digitise music, we need $f_s \geq 2 \times 20 = 40\,\text{kHz}$. This is why audio CDs use $f_s = 44.1\,\text{kHz}$ — slightly above the Nyquist rate, leaving a safety margin.

Step 2: Quantisation. Each sampled value is rounded to the nearest of a finite set of allowed levels.

DEFINITION **Quantisation**

Quantisation is the process of approximating each sampled value by the nearest of a set of discrete levels. If each sample is coded on n **bits**, the number of levels is

$$N = 2^n$$

and the quantisation step (resolution) over a voltage range $V_{max} - V_{min}$ is

$$q = \frac{V_{max} - V_{min}}{2^n}.$$

Step 3: Coding. Each level is given a binary number (a "word") of n bits. The output of the ADC is therefore a sequence of binary words — a digital signal.

METHOD **Number of quantisation levels, resolution and bit rate**

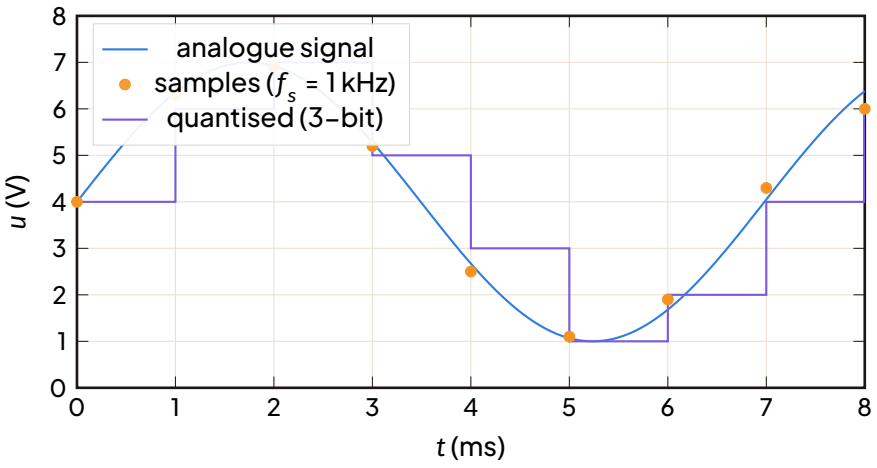
1. Identify the number of bits n per sample.
2. Compute the number of levels: $N = 2^n$.
3. Compute the step: $q = \frac{V_{max} - V_{min}}{N}$.
4. The bit rate of the digitised signal is $D = n \times f_s$ (bits per second, bit s^{-1}).

WORKED EXAMPLE 6 **Digitising a microphone signal**

A microphone produces a voltage between 0 and 8 V. It is digitised with a 3-bit ADC at a sampling frequency $f_s = 8$ kHz.
(a) How many quantisation levels are there? (b) Find the quantisation step. (c) Find the bit rate. (d) What binary word represents a sample of value 5.6 V?

SOLUTION

Solution.
(a) $N = 2^3 = 8$ levels.
(b) $q = \frac{8 - 0}{2^3} = 1$ V. The levels are 0, 1, 2, … , 7 V, coded 000, 001, … , 111.
(c) Bit rate: $D = nf_s = 3 \times 8000 = 2.4e4$ bit s^{-1} = 24 kbit s^{-1}.
(d) 5.6 V lies between levels 5 and 6; the nearest level is 6 V, coded **110**.



Sampling and 3-bit quantisation of an analogue signal. The continuous curve is sampled at regular intervals; each sample is rounded to the nearest of the $2^3 = 8$ allowed levels.

4.3 Digital-to-Analogue Conversion (DAC)

At the receiver, the reverse operation is needed: the stream of binary words must be turned back into a continuously varying voltage to drive a loudspeaker or a screen.

DEFINITION **Digital-to-analogue conversion**

A **DAC** reads each binary word and outputs the corresponding voltage level, holding it until the next word arrives. The raw output is a staircase signal; a low-pass **smoothing filter** then removes the sharp steps and restores a smooth analogue waveform.

The staircase is only an approximation of the original signal. The difference between the staircase and the true signal is the quantisation error; it is reduced by using more bits per sample (smaller step q) and a higher sampling frequency.

WARNING **Quantisation is irreversible**

Rounding to the nearest level destroys information: two different samples can be coded by the same word. No DAC can recover the lost detail. This is why the number of bits must be chosen before transmission.

4.4 Digital versus Analogue Transmission

	Analogue	Digital
Noise	Noise added along the line is amplified with the signal and accumulates.	**Repeaters** regenerate clean pulses: they only need to decide "0 or 1", so noise does not accumulate.
Errors	Impossible to detect or correct.	Extra parity/check bits allow **error detection and correction**.
Capacity	One signal per channel.	**Multiplexing**: many digital signals interleaved on one channel; **compression** reduces the bit rate needed.
Cost/complexity	Simple circuits.	Needs ADC/DAC and more **bandwidth** for the same message.

KEY POINTS **Why digital won**

Digital signals can be regenerated (noise removed at each repeater), protected (error-correcting codes), compressed and multiplexed. Their price: greater bandwidth and conversion electronics.

4.5 Bandwidth and Sidebands

Every channel (cable, radio link, fibre) can carry only a limited range of frequencies.

DEFINITION **Bandwidth**

The **bandwidth** B of a channel is the width of the range of frequencies it can transmit:

$$B = f_{\text{high}} - f_{\text{low}} \qquad \text{(unit: Hz).}$$

A signal whose spectrum is wider than the channel bandwidth is distorted.

Sidebands in AM. When a carrier f_c is amplitude-modulated by a sinusoidal signal of frequency f_m, the modulated wave is

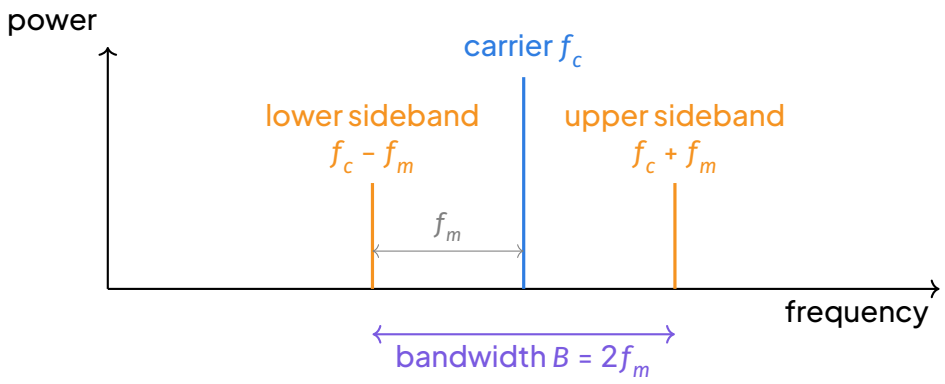
$$u(t) = U_c \cos(2\pi f_c t) + \frac{mU_c}{2}\cos\big(2\pi(f_c - f_m)t\big) + \frac{mU_c}{2}\cos\big(2\pi(f_c + f_m)t\big),$$

where m is the modulation index. The spectrum therefore contains three frequencies: the **carrier** f_c and two **sidebands** at $f_c - f_m$ and $f_c + f_m$. For a real message occupying frequencies up to $f_{m,\max}$, the sidebands extend from $f_c - f_{m,\max}$ to $f_c + f_{m,\max}$, so the AM signal occupies

$$\boxed{B_{\text{AM}} = 2\,f_{m,\max}}$$

WARNING **The classic trap**

The AM bandwidth counts both sidebands: it is $2f_m$, not f_m. If speech occupies up to 4.5 kHz, the AM channel needs $B = 2 \times 4.5 = 9$ kHz.



Frequency spectrum of an AM signal: carrier at f_c and two sidebands at $f_c \pm f_m$. The total bandwidth is $2f_m$. Most of the power sits in the carrier, which carries no information.

Power distribution. In AM, the carrier contains most of the transmitted power yet carries no information; only the sidebands do (for sinusoidal modulation of index m, each sideband carries a fraction $m^2/4$ of the carrier power). This is one weakness of AM. In **FM**, the information is coded in frequency deviations; the spectrum contains many side frequencies, and Carson's rule gives approximately

$$B_{\text{FM}} \approx 2(\Delta f + f_{m,\max}),$$

where Δf is the peak frequency deviation. Since Δf is large (e.g. 75 kHz for FM radio), FM needs a much **wider bandwidth** than AM — the price paid for its superior immunity to noise.

WORKED EXAMPLE 7 **AM channel bandwidth**

A radio station broadcasts on a carrier $f_c = 1.0$ MHz. The audio signal contains frequencies up to 5 kHz.
(a) Sketch the spectrum and give the frequencies of the extreme sideband components.
(b) Find the bandwidth of the channel. (c) How many such stations fit between 530 and 1600 kHz?

SOLUTION

Solution.
(a) Sidebands extend from $f_c - f_m = 1000 - 5 = 995$ kHz to $f_c + f_m = 1005$ kHz.
(b) $B = 2f_m = 2 \times 5 = 10$ kHz.

(c) Available band: 1600 − 530 = 1070 kHz. Number of channels: $N = \frac{1070}{10} = 107$ stations.

4.6 Satellite Communication

Microwaves travel in straight lines and are not reflected by the ionosphere, so Earth-to-Earth microwave links are limited by the horizon. The solution is to place a relay in the sky: a **communication satellite**.

PROPERTY **Geostationary orbit**

A **geostationary satellite** orbits in the equatorial plane, in the same direction as the Earth's rotation, with a period $T = 24$ h. Its altitude is about 36000 km above the equator (orbital radius ≈ 42000 km from the Earth's centre). It therefore appears fixed above one point of the equator, so ground antennas can point at it permanently. (The altitude follows from Kepler's third law: $T^2 = \frac{4\pi^2 r^3}{GM_T}$.)

EXEMPLE **Checking the geostationary radius**

With $T = 24 \times 3600 = 8.64e4$ s, $G = 6.67e-11$ N m^2 kg^{-2}, $M_T = 6.0e24$ kg:

$$r^3 = \frac{GM_T T^2}{4\pi^2} = \frac{6.67 \times 10^{-11} \times 6.0 \times 10^{24} \times (8.64 \times 10^4)^2}{39.5} \approx 7.6 \times 10^{22}\ \text{m}^3$$

so $r \approx 4.2e7$ m = 42000 km, i.e. an altitude of about 36000 km above the surface. ✓

Uplink and downlink. The ground station transmits to the satellite on the **uplink** frequency; the satellite amplifies the signal, shifts its frequency, and retransmits to Earth on the **downlink** frequency. Different frequencies (e.g. around 6 GHz up / 4 GHz down, or 14/12 GHz) are used so that the powerful uplink does not swamp the weak downlink received by the satellite.

Advantages: one satellite covers about a third of the globe — ideal for continental TV distribution, telephony and internet in regions without cables (vital for rural Cameroon). **Limitations:** the round trip of 2 × 36000 km at $c = 3.00e8$ m s^{-1} introduces a delay

$$\Delta t = \frac{2 \times 3.6 \times 10^7}{3.00 \times 10^8} \approx 0.24\ \text{s},$$

noticeable in telephone conversations; launching and maintaining satellites is costly; signals are weakened by rain at high frequencies.

4.7 Mobile Telephony: Cells and Base Stations

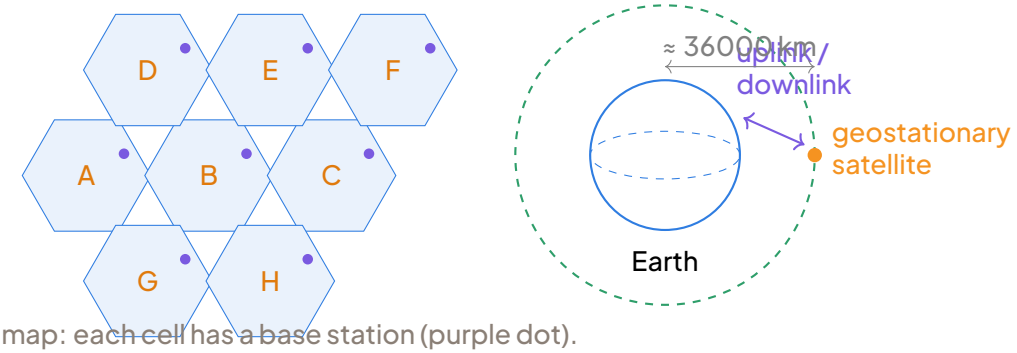
A mobile phone is a low-power radio transmitter: it cannot reach a whole country. The solution is the **cellular network**: the territory is divided into small **cells**, each served by a **base station**

(the masts you see along roads and on rooftops in Douala or Bamenda).

DEFINITION **Base station and its role**

A **base station** is a fixed radio transceiver that links the mobile handsets within its cell to the rest of the network (switching centres, other cells, the internet). Its roles are:

- to receive the weak signals from handsets and relay them into the network;
- to transmit network signals to the handsets;
- to manage **frequency reuse**: the same frequencies are reused in distant, non-adjacent cells without interference, multiplying the capacity of the network;
- to perform **handover**: as a user moves (e.g. in a taxi), the call is transferred seamlessly from one base station to the next.



Cellular map: each cell has a base station (purple dot). Frequencies of cell A are reused in cell F, far away.

Left: a region divided into hexagonal cells, each with a base station; frequencies are reused in distant cells. Right: a geostationary satellite fixed above the equator, exchanging uplink and downlink signals with a ground station.

4.8 A High-Bandwidth Channel: Optical Fibres (GCE link to optics)

For fixed networks, the champion channel is the **optical fibre**: a hair-thin glass fibre guides light pulses by **total internal reflection** at the core–cladding boundary (the phenomenon you met in geometrical optics: light strikes the interface at an angle greater than the critical angle, so it is trapped inside the core). Because light frequencies (~ 10^{14} Hz) are enormous, the available bandwidth is huge — a single fibre can carry millions of simultaneous phone calls or high-speed internet, with very low attenuation and total immunity to electromagnetic interference. Submarine fibre-optic cables landing on the Cameroonian coast connect the country to the global internet backbone.

DID YOU KNOW? **Did you know?**

Cameroon's international internet traffic arrives largely through submarine optical fibre cables landing near Douala, while domestic links combine fibre, microwave relays and satellite. The mobile phone in your pocket constantly negotiates with the nearest base station — and hands over to the next one as you travel, without you noticing.

WORKED EXAMPLE 8 **Bit rate of a digitised telephone channel (GCE style)**

A telephone voice channel is limited to $f_{max} = 3.4$ kHz. It is sampled at the Nyquist rate and coded on 8 bits per sample.
(a) Find the sampling frequency. (b) Find the bit rate of the digital channel. (c) How many such channels can a 2.048 Mbit s^{-1} digital link carry?

SOLUTION

Solution.
(a) Nyquist: $f_s = 2f_{max} = 2 \times 3.4 = 6.8$ kHz.
(b) $D = nf_s = 8 \times 6.8 \times 10^3 = 5.44e4$ bit $s^{-1} = 54.4$ kbit s^{-1}.
(c) $N = \dfrac{2.048 \times 10^6}{5.44 \times 10^4} \approx 37.6$, so 37 channels (whole number, rounded down).

EXERCISE 4 **Practice**

1. An audio signal occupies 20 Hz to 15 kHz. It is digitised on 16 bits. (a) Give the minimum sampling frequency. (b) How many quantisation levels are available? (c) Calculate the bit rate.
2. An AM transmitter uses a carrier of 800 kHz and modulating frequencies up to 4 kHz. State the frequencies present in the spectrum and the channel bandwidth.
3. Explain, in three sentences, why digital repeaters give better long-distance quality than analogue amplifiers.
4. A geostationary satellite is 36000 km above the Earth's surface. Calculate the total time delay for a signal going Earth → satellite → Earth.

SOLUTION **Exercise 1**

1. (a) $f_s \geq 2 \times 15 = 30$ kHz. (b) $N = 2^{16} = 65536$ levels. (c) $D = 16 \times 30\,000 = 4.8e5$ bit $s^{-1} = 480$ kbit s^{-1}.

2. Carrier 800 kHz; sidebands from 800 − 4 = 796 kHz to 800 + 4 = 804 kHz; bandwidth $B = 2f_m = 8\,\text{kHz}$.
3. Analogue amplifiers amplify the noise together with the signal, so noise accumulates along the line. Digital repeaters only decide whether each pulse is a 0 or a 1, then re-transmit a clean, full-size pulse. Errors can additionally be detected and corrected with check bits, so quality is preserved over any distance.
4. $\Delta t = \dfrac{2 \times 3.6 \times 10^7\,\text{m}}{3.00 \times 10^8\,\text{m s}^{-1}} = 0.24\,\text{s}$.

KEY POINTS **Section summary**

- An analogue signal is continuous in time and value; a digital signal takes discrete values coded in bits.
- ADC = sampling ($f_s \geq 2f_{max}$, Nyquist) + quantisation (2^n levels, step $q = \Delta V / 2^n$) + binary coding. Bit rate $D = nf_s$.
- DAC = binary words → staircase voltage → smoothing filter.
- Digital transmission: regeneration by repeaters, error correction, compression, multiplexing — at the cost of bandwidth and converters.
- Bandwidth $B = f_{high} - f_{low}$; AM spectrum = carrier + two sidebands at $f_c \pm f_m$, so $B_{AM} = 2f_{m,max}$; FM is wider, $B_{FM} \approx 2(\Delta f + f_{m,max})$.
- Geostationary satellite: equatorial orbit, $T = 24\,\text{h}$, altitude ≈ 36000 km; uplink and downlink on different frequencies; delay ≈ 0.24 s.
- Cellular telephony: cells, base stations (link handsets to network, frequency reuse, handover). Optical fibres: total internal reflection, enormous bandwidth.

5 INTEGRATION ACTIVITY

5.1 Aims of the activity

By the end of this activity, you should be able to:

- **analyse** a real communication problem and identify the physical constraints (distance, terrain, frequency allocation);
- **select** an appropriate propagation mode (ground wave, sky wave, space wave) for a given link and justify the choice;
- **calculate** the dimensions of a half-wave dipole aerial for a given carrier frequency;
- **design** a simple *LC* tuning circuit and determine the range of capacitance needed to cover a given band;
- **compare** AM and FM and defend an engineering choice using their advantages and disadvantages;
- **compute** the bandwidth of an AM signal and the number of channels fitting in a given allocation;
- **propose** a long-distance relay solution (satellite link or network of base stations) and justify it;
- **communicate** scientific results in the form of a clear, argued one-page engineering brief.

5.2 Situation

Radio Menchum Community Project.

The NGO Ndawara Development Trust has obtained a licence to operate a community radio station, **Radio Menchum**, with its studio and transmitter located on a hill near **Bamenda** (North-West Region). The station must serve farming villages up to 80 km away across the hilly, forested terrain of the region. In addition, recorded programmes must be relayed every evening to a partner station in **Yaoundé**, about 280 km away, for rebroadcast in the Centre Region.

The engineering team (your group) is given the following data and constraints:

Item	Data / constraint
Local broadcast range	80 km, hilly terrain, no line of sight for many villages
Available bands	MF band around 1.0 MHz (AM), or VHF FM band 88–108 MHz
Relay link Bamenda–Yaoundé	280 km, mountainous terrain, no line of sight
Highest audio frequency	$f_{m,\max} = 5.0$ kHz
AM allocation in the region	200 kHz of spectrum reserved for community AM stations
Tuning circuit inductor	$L = 0.50$ µH available in the workshop
Speed of electromagnetic waves	$c = 3.00e8$ m s^{-1}

The Trust asks each group to produce a **one-page engineering brief** containing: the chosen frequency bands with justification of the propagation mode for each link, the design of the transmitting aerial, the design of the receiver tuning circuit, the choice of modulation with a argued defence, the bandwidth calculations, and a diagram of the complete communication chain (transmitter → channel → receiver, plus the relay to Yaoundé).

5.3 Tasks

1. **Propagation modes.** For the local Bamenda–villages link (80 km, hilly), explain why a space wave at VHF is problematic without line of sight, and discuss the two candidate solutions: an MF ground wave (AM) or a VHF FM transmitter placed on a high hill. State which propagation mode each uses.

2. **Transmitting aerial.** The team finally decides to broadcast locally in FM at the carrier frequency f_c = 98.0 MHz. Calculate the wavelength and the length of a half-wave dipole transmitting aerial.

3. **Tuning circuit.** A listener's receiver uses a parallel LC circuit with L = 0.50 µH. Using $f_0 = \dfrac{1}{2\pi\sqrt{LC}}$, calculate the capacitance needed to tune to 88 MHz and to 108 MHz, and hence state the range of the variable capacitor. Sketch the resonance curve and indicate on it the tuning at 98.0 MHz.

4. **Choice of modulation.** For the local broadcast, choose between AM and FM. Give at least two advantages of your choice and one disadvantage, referring to sound quality, noise immunity, bandwidth and cost of receivers.

5. **Bandwidth.** Suppose instead that the station had used AM with a highest audio frequency of 5.0 kHz.

   (a) Calculate the bandwidth of the AM signal.

   (b) How many such stations can fit, without overlap, in the 200 kHz allocation?

6. **Relay to Yaoundé.** Explain why neither a ground wave nor a direct space wave is suitable for the 280 km Bamenda–Yaoundé link. Propose two realistic solutions (a chain of base stations / microwave repeaters on hilltops, or a satellite link) and give one advantage and one disadvantage of each.

7. **Engineering brief.** Assemble your results into a one-page brief including a labelled block diagram of the complete system: microphone → audio amplifier → modulator → carrier oscillator → power amplifier → transmitting aerial → channel → receiving aerial → tuning circuit → demodulator → loudspeaker, plus the relay link.

5.4 Model answer

1. Propagation modes for the local link.

At VHF (~ 100 MHz), radio waves behave as a **space wave**: they travel essentially in straight lines and are blocked by hills. Since many villages have no line of sight to Bamenda, a VHF transmitter in the town itself would leave shadow zones. Two solutions exist:

- an **MF ground wave** (around 1 MHz): the wave follows the curvature of the Earth and diffracts around obstacles, covering 80 km of hilly terrain reliably, but only AM is available there and the audio quality is limited;
- a **VHF FM space wave** with the transmitting aerial on a high hill (e.g. the Bamenda escarpment): the height of the aerial extends the line-of-sight horizon to several tens of kilometres, so most villages within 80 km can be reached, with the superior quality of FM.

The team chooses the second solution: **FM at** 98.0 MHz**, space-wave propagation, hill-top site**.

2. Half-wave dipole.

$$\lambda = \frac{c}{f_c} = \frac{3.00 \times 10^8}{98.0 \times 10^6} = 3.06\ \text{m}.$$

The dipole length is

$$\ell = \frac{\lambda}{2} = \frac{3.06}{2} \approx 1.53\ \text{m},$$

i.e. two rods of about 0.77 m each, mounted horizontally for horizontal polarisation.

3. Tuning circuit.

From $f_0 = \dfrac{1}{2\pi\sqrt{LC}}$ we get $C = \dfrac{1}{4\pi^2 L f_0^2}$.

$$C_{88} = \frac{1}{4\pi^2(0.50 \times 10^{-6})(88 \times 10^6)^2} \approx 6.5\ \text{pF}, \qquad C_{108} = \frac{1}{4\pi^2(0.50 \times 10^{-6})(108 \times 10^6)^2} \approx 4.3\ \text{pF}.$$

The variable capacitor must cover approximately $\boxed{4.3\ \text{pF to } 6.5\ \text{pF}}$. On the resonance curve (current versus frequency, peaked at f_0), tuning to 98.0 MHz corresponds to $C \approx 5.3$ pF, placing the peak of the curve on the desired carrier so that Radio Menchum is selected and neighbouring stations are rejected.

4. Choice of modulation: FM.

- Advantages of FM: much better immunity to noise and electrical interference (lightning, motorbikes, generators), giving clear sound; better audio quality with a wider audio bandwidth, suitable for music.
- Disadvantage: FM requires a much larger bandwidth per station and more complex, slightly more expensive receivers; also its space-wave propagation limits coverage to (quasi) line of sight, which is why the hill-top site is essential.

AM would offer simpler, cheaper receivers and ground-wave coverage, but the noisy tropical environment (thunderstorms) would severely degrade reception. **FM is retained.**

5. AM bandwidth (hypothetical AM option).

1. An AM signal occupies two sidebands: $B = 2f_{m,\max} = 2 \times 5.0 = 10$ kHz.
2. Number of stations in 200 kHz: $N = \dfrac{200}{10} = \boxed{20}$ stations (in practice slightly fewer, to leave guard bands).

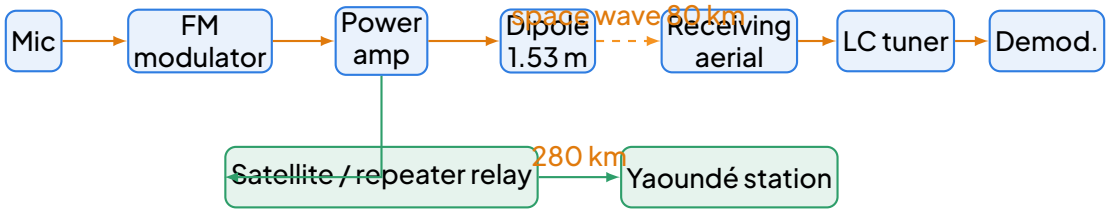
6. Relay to Yaoundé.

At 280 km over mountains, the ground wave is far too attenuated and there is no line of sight for a space wave; sky-wave reflection is unreliable for a daily high-quality programme feed. Two solutions:

- **Chain of base stations / microwave repeaters** on hilltops every 30–50 km: each base station receives, amplifies and retransmits the signal. Advantage: fully under local control, moderate running cost. Disadvantage: several sites to build, power and maintain in difficult terrain.
- **Satellite link:** the programme is sent up to a geostationary satellite (uplink) and received in Yaoundé by a dish (downlink). Advantage: a single hop covers the whole distance with excellent quality. Disadvantage: high cost of satellite capacity and of the uplink equipment.

The team proposes the satellite link for reliability, with a single hill-top microwave repeater as a backup.

7. Engineering brief (summary diagram).



Block diagram of the Radio Menchum system: FM broadcast chain and relay to Yaoundé.

WARNING **Marking tips**

Every numerical answer must carry its unit and a justification; a choice (band, modulation, relay) earns full marks only when defended by at least one physical argument drawn from the course (propagation mode, noise immunity, bandwidth, line of sight). A diagram without labels, or calculations without a concluding sentence, loses marks.

6 METHODS AND WORKED EXAMPLES

7 COMMUNICATION

7.1 Representing Information and Radio Wave Propagation

DEFINITION **Analogue and Digital Signals**

Analogue Signal: A continuous signal in both time and amplitude. It can take any value within a range (e.g., voice, temperature, natural light intensity). Digital Signal: A discrete signal in both time and amplitude. It is represented by a sequence of binary digits (bits), typically two voltage levels representing 0 and 1.

KEY POINTS **Comparison: Analogue vs Digital Representation**

Characteristic	Analogue	Digital
Resolution	Infinite (continuous)	Finite (quantisation steps)
Noise Immunity	Poor (noise adds directly)	High (regeneration, error correction)
Processing	Simple (amplifiers, filters)	Complex (requires ADC/DAC, DSP)
Storage/Transmission	Degrades with copies/distance	Perfect copies, encryption possible
Bandwidth Efficiency	High (baseband)	Lower (requires higher BW for same info)

PROPERTY **Radio Wave Propagation Modes**

The propagation mode depends primarily on frequency and distance.

1. Ground (Surface) Wave: LF/MF (< 3 MHz). Wave follows Earth's curvature via diffraction. Attenuation increases with frequency and ground conductivity (sea > land). Range ∝ Transmitter Power. Uses: AM Radio, Maritime/Naval comms, Time signals.
2. Sky Wave (Ionospheric): HF (3–30 MHz). Wave refracted (bent) back to Earth by

ionised layers (D, E, F1, F2). Enables intercontinental communication.

- Day: D-layer absorbs lower HF; F-layers refract higher HF.
- Night: D-layer vanishes; lower HF refract well off E/F layers → longer range but fading/multipath.
- Critical Frequency (f_c): Highest frequency refracted vertically. $f_c = 9\sqrt{N_{max}}$ (N_{max} = max electron density m^{-3}).
- Maximum Usable Frequency (MUF): MUF = $f_c \sec \phi_i$ (ϕ_i = angle of incidence).
- Skip Zone: Region between ground wave limit and first sky wave return → no reception.

3. Space Wave (Line-of-Sight - LOS): VHF/UHF/SHF (> 30 MHz). Travels in straight lines. Range limited by radio horizon.
   - Radio Horizon Distance: $d_{horizon} \approx \sqrt{2kR_E h} \approx 3.57\ \text{km}\sqrt{h(\text{m})}$ (standard $k = 1$) or $\approx 4.12\ \text{km}\sqrt{h(\text{m})}$ (effective Earth radius factor $k = 4/3$ accounting for standard atmospheric refraction). GCE typically uses $3.57\ \text{km}\sqrt{h}$.
   - Max LOS Range: $d_{max} \approx 3.57\ \text{km}\left(\sqrt{h_T} + \sqrt{h_R}\right)$.
   - Uses: FM/TV Broadcast, Mobile Phones, Satellite, Radar, Microwave Links.

METHOD **Selecting Propagation Mode for a Link**

1. Identify Frequency Band: LF/MF → Ground; HF → Sky; VHF+ → Space.
2. Calculate LOS Horizon: $d_{max} = 3.57(\sqrt{h_T} + \sqrt{h_R})$ km (h in m).
3. Check Distance vs Horizon: If $d > d_{max}$ at VHF+, link impossible without relay/satellite.
4. Assess Ionospheric Conditions (HF): Time of day, solar cycle, MUF vs Operating Frequency.
5. Ground Wave Range (MF): Estimate via empirical curves (Millington) or $E \propto P_t/d^2$ (simplified); conductivity crucial.

WORKED EXAMPLE 9 **Radio Horizon and Propagation Mode Selection**

A transmitting aerial is mounted on a mast 150 m high. The receiving aerial is 20 m above ground. The carrier frequency is 100 MHz.

1. Calculate the maximum line-of-sight range for space wave propagation.

2. Explain why ground wave propagation is ineffective at this frequency.
3. If the frequency were changed to 1 MHz (Medium Wave), which propagation mode would dominate for a distance of 200 km? Justify.

SOLUTION

Detailed Solution

1. Space Wave Range (Line-of-Sight): The maximum distance d_{max} is the sum of the horizon distances of transmitter (h_T) and receiver (h_R).

$$d_{max} \approx 3.57\left(\sqrt{h_T} + \sqrt{h_R}\right) \quad \text{with } h \text{ in metres, } d \text{ in km.}$$

$h_T = 150$ m, $h_R = 20$ m.

$$\sqrt{150} \approx 12.25, \quad \sqrt{20} \approx 4.47$$

$$d_{max} \approx 3.57 \times (12.25 + 4.47) = 3.57 \times 16.72 \approx 59.7 \text{ km.}$$

Answer: Maximum LOS range ≈ 60 km.

2. Ground Wave at 100 MHz: Ground wave attenuation increases sharply with frequency due to dielectric losses in the Earth and diffraction losses. At VHF (100 MHz), the ground wave is heavily attenuated within a few kilometres. It is negligible beyond the horizon. Space wave dominates at VHF/UHF.
3. Propagation at 1 MHz (MF) over 200 km: At 1 MHz (Medium Frequency), ground wave propagation is effective (range up to few hundred km over land/sea depending on power/conductivity). However, for 200 km:
   - Daytime: D-layer absorbs sky wave; Ground Wave is the reliable mode.
   - Nighttime: D-layer disappears; Sky wave (refraction from E/F layers) becomes very strong and often dominates, but suffers from fading/multipath interference.

   Conclusion: For a stable 200 km link, Ground Wave is the primary intended mode (especially daytime), though Sky Wave may interfere at night. Space wave is impossible (horizon ≈ 60 km).

7.2 Aerials and the Tuning Circuit

DEFINITION **Aerial (Antenna) Parameters**

- Radiation Pattern: 3D plot of radiated power vs direction. Isotropic (theoretical, equal all directions), Omni-directional (e.g., vertical monopole), Directional (e.g., Yagi, Dish).

- Gain (G): Ratio of power density in preferred direction to that of an isotropic radiator with same input power. Unit: dBi (ref: isotropic) or dBd (ref: half-wave dipole, 0 dBd = 2.15 dBi).
- Effective Length (h_{eff}): $V_{\text{oc}} = E \cdot h_{\text{eff}}$ (Open-circuit voltage induced by field strength E).
- Aperture (A_e): Effective capture area. $A_e = \frac{G\lambda^2}{4\pi}$.
- Polarisation: Orientation of $\vec{E}$-field (Vertical, Horizontal, Circular). Tx and Rx must match for max signal (cross-pol loss $\to \infty$ ideally).
- Common Types: Half-wave Dipole ($L = \lambda/2$, $G \approx 2.15$ dBi), Quarter-wave Monopole (needs ground plane), Yagi-Uda (directional, high gain), Parabolic Dish (very high gain, SHF/Satellite), Loop Aerial (direction finding).

PROPERTY **Parallel LC Tuning Circuit (Resonance)**

Used in RF amplifiers and receivers to select desired frequency f_0.

- Circuit: Inductor L (with series loss resistance r) in parallel with Capacitor C (assumed ideal).
- Resonance Frequency: $f_0 = \frac{1}{2\pi\sqrt{LC}}$. Derivation (Examinable): Admittance $Y = \frac{1}{r + j\omega L} + j\omega C$. Set $\Im(Y) = 0 \implies \omega_0 C = \frac{\omega_0 L}{r^2 + \omega_0^2 L^2} \approx \frac{1}{\omega_0 L}$ (for $Q \gg 1$).
- Dynamic Resistance (R_{dyn}): Impedance at resonance (purely resistive).

$$R_{\text{dyn}} = \frac{L}{rC} = Q\omega_0 L = \frac{Q}{\omega_0 C}$$

- Q-Factor (Quality Factor):

$$Q = \frac{\omega_0 L}{r} = \frac{1}{\omega_0 C r} = \frac{f_0}{\Delta f}$$

  where Δf is the 3 dB bandwidth (frequency difference between half-power points).
- Resonance Curve: Plot of $|Z|$ or I vs f. Peak at f_0, width Δf at $1/\sqrt{2}$ of peak. High $Q \to$ Sharp peak $\to$ High Selectivity.
- Selectivity: Ability to reject adjacent channels. $\frac{V_{\text{out}}(f_0 \pm \delta f)}{V_{\text{out}}(f_0)} \approx \frac{1}{\sqrt{1 + Q^2\left(\frac{2\delta f}{f_0}\right)^2}}$ (for $\delta f \ll f_0$).

METHOD **Tuning Circuit Analysis: Resonance, Q-Factor and Selectivity**

1. Identify Configuration: Parallel LC (practical: L with series r, C ideal). At resonance, $Z_{max} = R_{dyn}$.
2. Calculate f_0: $f_0 = 1/(2\pi\sqrt{LC})$.
3. Calculate Q: $Q = \omega_0 L/r$ (series r given) or $Q = R_p/(\omega_0 L)$ (parallel R_p given). Check which resistance is given!
4. Find Bandwidth: $\Delta f = f_0/Q$.
5. Assess Selectivity: Compute attenuation at adjacent channel offset δf using approximation formula or exact impedance formula.

WORKED EXAMPLE 10 **Tuning Circuit Design and Selectivity**

A radio receiver uses a parallel LC tuning circuit. The inductor has inductance $L = 200\ \mu H$ and series resistance $r = 10\ \Omega$. The capacitor is variable.

1. Calculate the capacitance required to tune to a station at 95 MHz (FM band).
2. Determine the Q-factor of the circuit at this frequency.
3. Calculate the 3 dB bandwidth Δf.
4. If the adjacent channel is 200 kHz away (standard FM spacing), will this circuit provide adequate selectivity? (Assume adequate if adjacent signal is attenuated by > 20 dB relative to resonance).

SOLUTION

Detailed Solution

1. Resonance Capacitance: $f_0 = 95\ \text{MHz} = 95 \times 10^6\ \text{Hz}$.

$$f_0 = \frac{1}{2\pi\sqrt{LC}} \implies C = \frac{1}{(2\pi f_0)^2 L}$$

$$(2\pi f_0)^2 = (2\pi \times 95 \times 10^6)^2 \approx (5.969 \times 10^8)^2 \approx 3.563 \times 10^{17}\ \text{s}^{-2}$$

$$C = \frac{1}{3.563 \times 10^{17} \times 200 \times 10^{-6}} = \frac{1}{7.126 \times 10^{13}} \approx 1.40 \times 10^{-14}\ \text{F} = 14.0\ \text{pF}.$$

2. Q-Factor: $\omega_0 = 2\pi f_0 = 2\pi \times 95 \times 10^6 \approx 5.97 \times 10^8\ \text{rad/s}$.

$$Q = \frac{\omega_0 L}{r} = \frac{5.97 \times 10^8 \times 200 \times 10^{-6}}{10} = \frac{1.194 \times 10^5}{10} = 11940.$$

Note: This Q is extremely high for a discrete component at VHF; typical values are lower due to stray capacitance/skin effect. But we use given data.

3. Bandwidth:
$$\Delta f = \frac{f_0}{Q} = \frac{95 \times 10^6}{11940} \approx 7950 \text{ Hz} \approx 8.0 \text{ kHz}.$$

4. Selectivity Check: Channel spacing = 200 kHz. Offset from resonance $f_{adj} = f_0 \pm 200$ kHz. Relative frequency deviation: $\frac{\Delta f_{offset}}{f_0} = \frac{200 \times 10^3}{95 \times 10^6} \approx 2.105 \times 10^{-3}$. For a parallel resonant circuit, the impedance ratio (voltage gain ratio) at offset δf is approx:
$$\frac{V_{out}(\delta f)}{V_{out}(f_0)} \approx \frac{1}{\sqrt{1 + Q^2\left(\frac{2\delta f}{f_0}\right)^2}}$$
(Using approximation for small δf: $\frac{\omega L - 1/\omega C}{r} \approx Q\frac{2\delta f}{f_0}$).
$$Q\frac{2\delta f}{f_0} = 11940 \times 2 \times 2.105 \times 10^{-3} \approx 50.3$$
Attenuation $\approx 20 \log_{10}(50.3) \approx 34$ dB. Since 34 dB > 20 dB, the selectivity is more than adequate. Comment: In practice, FM receivers use IF filters (ceramic/SAW) for channel selection, not just the RF tuning circuit, because tracking a high-Q circuit over the band is difficult.

7.3 Modulation, Demodulation, AM and FM

DEFINITION **Modulation**

Process of varying a parameter of a high-frequency carrier wave ($c(t) = E_c \sin \omega_c t$) in accordance with a low-frequency modulating signal ($m(t) = E_m \sin \omega_m t$).

- Amplitude Modulation (AM): Carrier amplitude varies. $v_{AM}(t) = E_c[1 + m \sin \omega_m t] \sin \omega_c t$.
- Frequency Modulation (FM): Carrier frequency varies. $v_{FM}(t) = E_c \sin[\omega_c t + \beta \sin \omega_m t]$.
- Phase Modulation (PM): Carrier phase varies. (Related to FM).

Demodulation: Reverse process: recovering $m(t)$ from modulated wave.

PROPERTY **Amplitude Modulation (AM) – Parameters, Spectrum, Power**

- Modulation Index (Depth): $m = \frac{E_m}{E_c} = \frac{V_{max} - V_{min}}{V_{max} + V_{min}}$. Condition for no envelope distortion: $0 \le m \le 1$. Over-modulation ($m > 1$) causes phase reversal/distortion.

- AM Wave Equation (Single Tone):

$$v(t) = E_c \sin \omega_c t + \frac{mE_c}{2} \cos(\omega_c - \omega_m)t - \frac{mE_c}{2} \cos(\omega_c + \omega_m)t$$

- Frequency Spectrum (DSB-FC – Double Sideband Full Carrier):

Component	Frequency	Amplitude (Peak)
Carrier	f_c	E_c
Lower Sideband (LSB)	$f_c - f_m$	$mE_c/2$
Upper Sideband (USB)	$f_c + f_m$	$mE_c/2$

- Bandwidth: $BW = 2f_m(\text{max})$.
- Power (Load R): $P_c = \frac{E_c^2}{2R}$ (Carrier Power). $P_{SB} = \frac{(mE_c/2)^2}{2R} = \frac{m^2}{4}P_c$ (Power in each sideband). $P_{total} = P_c + 2P_{SB} = P_c\left(1 + \frac{m^2}{2}\right)$.
- Efficiency: $\eta = \frac{P_{sidebands}}{P_{total}} = \frac{m^2/2}{1 + m^2/2}$. Max $\eta = 33.3\%$ at $m = 1$.
- Variants: DSB-SC (Suppressed Carrier, 50% efficiency, needs coherent detection), SSB (Single Sideband, 100% info in half BW, complex).

WORKED EXAMPLE 11 **AM Power and Bandwidth Calculations**

An AM transmitter radiates 10 kW when the carrier is unmodulated. The carrier frequency is 1 MHz. The modulating signal is a single tone at 5 kHz.

1. Determine the carrier power P_c.
2. If the modulation depth is 80%, calculate the total transmitted power.
3. Calculate the power in each sideband.
4. Determine the transmission bandwidth.
5. If the transmitter is limited to 15 kW total power, what is the maximum modulation index possible?

SOLUTION

Detailed Solution

1. Carrier Power: Unmodulated power = Carrier power $P_c = 10$ kW.

2. Total Power at $m = 0.8$: $P_T = P_c\left(1 + \frac{m^2}{2}\right) = 10\left(1 + \frac{0.8^2}{2}\right) = 10\left(1 + \frac{0.64}{2}\right) = 10(1+0.32) = 13.2\,\text{kW}$.
3. Sideband Power (each): $P_{SB} = \frac{m^2}{4}P_c = \frac{0.64}{4} \times 10 = 0.16 \times 10 = 1.6\,\text{kW}$. Check: Total SB power = $2 \times 1.6 = 3.2\,\text{kW}$. $P_T = 10 + 3.2 = 13.2\,\text{kW}$. Correct.
4. Bandwidth: $BW = 2f_m = 2 \times 5\,\text{kHz} = 10\,\text{kHz}$. Spectrum occupies 995 kHz to 1005 kHz.
5. Maximum Modulation Index for $P_{T,\text{max}} = 15\,\text{kW}$: $15 = 10\left(1 + \frac{m_{\text{max}}^2}{2}\right) \Longrightarrow 1.5 = 1 + \frac{m_{\text{max}}^2}{2} \Longrightarrow 0.5 = \frac{m_{\text{max}}^2}{2} \Longrightarrow m_{\text{max}}^2 = 1 \Longrightarrow m_{\text{max}} = 1\,(100\%)$. The transmitter can handle full modulation without exceeding 15 kW.

PROPERTY **Frequency Modulation (FM) – Parameters, Spectrum, Bandwidth**

- Instantaneous Frequency: $f_i(t) = f_c + \Delta f \sin \omega_m t$. Δf = Peak Frequency Deviation (Hz).
- Modulation Index: $\beta = \frac{\Delta f}{f_m}$. β can be > 1 (unlike AM).
- FM Wave (Single Tone): $v(t) = E_c \sin(\omega_c t + \beta \sin \omega_m t)$.
- Spectrum: Infinite sidebands at $f_c \pm nf_m$ ($n = 0, 1, 2\ldots$). Amplitudes $E_c J_n(\beta)$ where J_n are Bessel functions of first kind. $J_0(\beta)$ = Carrier amplitude factor.
- Carson's Rule (Approximate Bandwidth for > 98% power):

$$BW \approx 2(\Delta f + f_m) = 2f_m(\beta + 1)$$

- Narrowband FM (NBFM): $\beta \ll 1$ (typically < 0.3). $BW \approx 2f_m$ (similar to AM). Used in 2-way radio (walkie-talkies, $\Delta f = 2.5$ kHz or 5 kHz).
- Wideband FM (WBFM): $\beta > 1$. Broadcast FM: $\Delta f = 75$ kHz, $f_m(\text{max}) = 15$ kHz $\Rightarrow \beta = 5$, $BW \approx 180$ kHz. High fidelity, noise immunity.
- Pre-emphasis / De-emphasis: High frequencies boosted at Tx (Pre-emphasis, time constant $\tau = 50\,\mu$s or $75\,\mu$s) and cut at Rx (De-emphasis) to improve SNR for high audio frequencies (triangular noise spectrum in FM).

WORKED EXAMPLE 12 **FM Bandwidth and Modulation Index Analysis**

An FM broadcast transmitter operates at 98 MHz with a maximum frequency deviation $\Delta f = 75$ kHz. The audio modulating signal ranges from 50 Hz to 15 kHz.

1. Calculate the modulation index β for the maximum modulating frequency (15 kHz) and

for the minimum (50 Hz).

2. Estimate the transmission bandwidth using Carson's Rule.
3. The transmitter is received by a receiver with an IF bandwidth of 200 kHz. Is this sufficient? Explain.
4. Compare the bandwidth required for this FM signal with an AM signal carrying the same audio bandwidth (15 kHz).

SOLUTION

Detailed Solution

1. Modulation Index β: $\beta = \frac{\Delta f}{f_m}$. $\Delta f = 75$ kHz (constant for a given transmitter drive level).
   - At $f_m = 15$ kHz: $\beta_{max} = \frac{75}{15} = \mathbf{5}$.
   - At $f_m = 50$ Hz: $\beta_{min} = \frac{75000}{50} = \mathbf{1500}$.

   Note: β varies inversely with f_m. Low frequencies produce very high β, generating many significant sidebands.
2. Carson's Rule Bandwidth: Use maximum modulating frequency f_m(max) = 15 kHz for bandwidth estimation. $BW \approx 2(\Delta f + f_m) = 2(75 + 15) = 180$ kHz. (Using Bessel table for $\beta = 5$: Significant sidebands up to $n \approx \beta + 1 = 6$. $BW = 2 \times 6 \times 15 = 180$ kHz. Matches perfectly).
3. Receiver IF Bandwidth Check: Required BW ≈ 180 kHz. Receiver IF BW = 200 kHz. Yes, it is sufficient. The IF bandwidth is slightly wider than Carson's bandwidth, allowing passage of ≈ 98% of signal power without excessive distortion, while rejecting adjacent channels (spaced 200 kHz apart in many regions).
4. Comparison with AM: AM Bandwidth for f_m(max) = 15 kHz: $BW_{AM} = 2 \times 15 = 30$ kHz. FM Bandwidth (Carson) = 180 kHz. FM requires 6 times the bandwidth of AM for the same baseband signal. This is the trade-off for FM's superior noise performance and fidelity.

KEY POINTS **Comparison: AM vs FM (GCE Exam Standard)**

Feature	AM (DSB-FC)	FM (WBFM)
Bandwidth	Narrow ($2f_m$)	Wide ($2(\Delta f + f_m)$)
Noise Immunity	Poor (amplitude noise adds directly)	Excellent (Limiter rejects AM noise)
Fidelity	Limited by narrow BW	High (wide BW allows 15 kHz audio)
Receiver Complexity	Simple (Envelope Detector)	Complex (Limiter + Discriminator/PLL)
Capture Effect	No (strongest adds)	Yes (stronger signal captures receiver)
Power Efficiency	Low (Carrier wastes > 50% power)	High (Constant amplitude, Class C amps)
Propagation	Ground/Sky wave (MF/HF)	Space wave only (VHF+)

METHOD **Demodulation Techniques**

1. AM Demodulation – Envelope Detector:
   - Circuit: Diode (rectifier) + Parallel $R_L C$ load.
   - Time Constant Condition: $\frac{1}{\omega_c} \ll R_L C \ll \frac{1}{\omega_m(\max)}$.
   - Too small RC: Carrier ripple on output. Too large RC: Diagonal clipping (failure to follow envelope decay when m high).
   - Output: $v_{out}(t) \propto E_c(1 + m \sin \omega_m t) \xrightarrow{\text{DC Block}}$ Audio.
2. FM Demodulation – Frequency Discriminators: Convert $\Delta f \rightarrow \Delta V \rightarrow$ Envelope Detect.
   - Slope Detector: Tuned circuit offset from f_c. Operates on slope of resonance curve. Simple but non-linear, poor linearity.
   - Foster-Seeley Discriminator: Centre-tapped transformer + two diodes. Output ∝ frequency deviation. Symmetrical S-curve. Good linearity. Requires Limiter before it (sensitive to AM).
   - Ratio Detector: Similar topology but with tertiary winding/long τ capacitor giving inherent limiting (AM rejection). No separate limiter strictly needed. Wider capture range.
   - Phase Locked Loop (PLL) – Modern Standard: VCO locks to input frequency. Control voltage ∝ frequency deviation → Demodulated output. Excellent linearity, noise performance, integrable (IC).

3. S-Curve: Plot of V_{out} vs f_{in}. Linear region = usable range. Slope = Sensitivity (V/Hz). Saturation limits = Dynamic Range ($\pm\Delta f$).

WORKED EXAMPLE 13 **AM Envelope Detector Design and FM Discriminator Characteristics**

An AM signal with carrier frequency 1 MHz and maximum modulating frequency 5 kHz (modulation index $m = 0.8$) is to be demodulated using a simple envelope detector. The load resistance $R_L = 10\,\text{k}\Omega$.

1. Determine the suitable range for the capacitor C to avoid both carrier ripple and diagonal clipping.
2. Select a standard capacitor value and verify.
3. For FM demodulation, sketch the ideal "S-curve" of a discriminator and label: Linear region, Sensitivity, Dynamic Range. Explain why a Limiter is essential before a Foster-Seeley discriminator but not strictly before a Ratio Detector.

SOLUTION

Detailed Solution

1. RC Time Constant Constraints: $\omega_c = 2\pi f_c = 2\pi \times 10^6 \approx 6.28 \times 10^6$ rad/s. $\omega_m(\text{max}) = 2\pi \times 5000 \approx 3.14 \times 10^4$ rad/s. Condition: $\frac{1}{\omega_c} \ll RC \ll \frac{1}{\omega_m}$. Lower bound (Carrier ripple): $RC \gg \frac{1}{6.28 \times 10^6} \approx 0.16\,\mu\text{s}$. Upper bound (Diagonal clipping): $RC \ll \frac{1}{3.14 \times 10^4} \approx 31.8\,\mu\text{s}$. Safe Range: $1\,\mu\text{s}$ to $10\,\mu\text{s}$ (Geometric mean $\approx 2.2\,\mu\text{s}$).
2. Component Selection: $R_L = 10\,\text{k}\Omega$. Choose $RC = 5\,\mu\text{s}$ (mid-range). $C = \frac{5 \times 10^{-6}}{10 \times 10^3} = 500$ pF. Standard value: 470 pF or 560 pF. Check $C = 470$ pF: $RC = 4.7\,\mu\text{s}$. $\frac{1}{\omega_c} = 0.16\,\mu\text{s} \ll 4.7\,\mu\text{s} \ll 31.8\,\mu\text{s} = \frac{1}{\omega_m}$. Condition satisfied.
3. FM Discriminator S-Curve and Limiter: Sketch Description: Graph of V_{out} (Vertical) vs f_{in} (Horizontal).
   - Shape: Inverted 'S' (or 'S' depending on polarity). Passes through origin (f_c, 0V).
   - Linear Region: Central portion where $V_{out} \propto (f_{in} - f_c)$. Demodulation occurs here.
   - Sensitivity (Slope): $S = dV/df$ (Volts/Hz or V/kHz) in linear region.
   - Dynamic Range / Peak Deviation Limit: Frequencies $f_c \pm \Delta f$ where curve starts to saturate/flatten. Beyond this, distortion occurs.

   Limiter Requirement:

   - Foster-Seeley: Output amplitude depends on input signal amplitude (AM component) AND frequency. Any amplitude noise (lightning, ignition, fading) appears

directly on the audio output. A Limiter (hard limiter/clipper) is mandatory to remove AM before discrimination.

- Ratio Detector: Circuit topology (diode charging a large capacitor C_3 with long time constant) creates a reference voltage proportional to the average signal amplitude. The output is taken as a ratio of voltages across the load resistors. This makes the output inherently independent of input amplitude variations (AM rejection built-in). No separate limiter stage is required, though one is often added for extra quieting.

7.4 Digital Conversion and Communication Channels

PROPERTY **Analogue-to-Digital Conversion (ADC)**

- Sampling (Nyquist Theorem): To reconstruct an analogue signal of maximum frequency f_{max} without aliasing, sampling frequency $f_s \geq 2f_{max}$. Practical systems use $f_s \geq 2.2f_{max}$ to allow for anti-aliasing filter roll-off.
- Quantisation: Mapping continuous amplitude to 2^N discrete levels. N = number of bits.
- Resolution (Step Size): $\Delta V = \frac{V_{FSR}}{2^N}$ where V_{FSR} is Full Scale Range.
- Quantisation Error: Maximum error = $\pm\Delta V/2$. RMS Quantisation Noise Voltage $V_{q,rms} = \frac{\Delta V}{\sqrt{12}}$.
- Signal-to-Quantisation-Noise Ratio (SQNR): For a full-scale sine wave ($V_{peak} = V_{FSR}/2$):

$$\text{SQNR}_{dB} \approx 6.02N + 1.76 \text{ dB}$$

  Rule of thumb: +1 bit ≈ +6 dB SQNR.
- ADC Types (Principles):
  - Flash (Parallel): Fastest, low resolution (< 8 bit), high power, $2^N - 1$ comparators.
  - Successive Approximation (SAR): Medium speed, 12–16 bit, 1 comparator + DAC + logic. Standard general purpose.
  - Sigma-Delta ($\Sigma\Delta$): Very high resolution (16–24+ bit), oversampling + noise shaping. Audio, instrumentation.
  - Dual Slope (Integrating): High accuracy, slow. Digital Multimeters (DMM).

PROPERTY **Digital-to-Analogue Conversion (DAC)**

- Reconstruction: Digital codes → Staircase voltage (Zero-Order Hold) → Reconstruction Filter (Low Pass, cutoff $f_c \approx f_s/2$) → Smoothed Analogue Output.
- Key Specs: Resolution (N bits), Settling Time, Glitch Energy (code transitions), Monotonicity.
- Common Architecture: R-2R Ladder (voltage mode) or Current Steering (high speed).

METHOD **ADC Specifications and Nyquist Rate**

1. Nyquist Frequency: $f_{max} < f_s/2$.
2. Resolution: $\Delta V = V_{FSR}/2^N$.
3. SQNR: $6.02N + 1.76$ dB (full-scale sine).
4. Anti-Aliasing Filter: Essential before ADC. Low-pass, cutoff $f_c \approx f_{max}$, stopband at $f_s - f_{max}$ with attenuation > SQNR to prevent aliases folding into baseband.
5. DAC Reconstruction: Zero-order hold → Sinc roll-off ($\sin x/x$) → Reconstruction Filter compensates.

WORKED EXAMPLE 14 **ADC Specifications and Nyquist Rate**

A digital audio system samples at 44.1 kHz (CD standard) using a 16-bit ADC. The input signal range is ±1 V.

1. What is the maximum audio frequency that can be recorded without aliasing?
2. Calculate the voltage resolution (step size) of the ADC.
3. Determine the theoretical Signal-to-Quantisation-Noise Ratio (SQNR) in dB.
4. If the system is upgraded to 24-bit at the same sampling rate, what is the new SQNR? By how many dB does it improve?
5. Explain why an anti-aliasing filter is essential before the ADC, and state its required cutoff frequency characteristic.

SOLUTION

Detailed Solution

1. Nyquist Frequency: $f_s = 44.1\,\text{kHz}$. $f_{\max} < \frac{f_s}{2} = \frac{44.1}{2} = 22.05\,\text{kHz}$. Answer: 22.05 kHz (Theoretical limit). Practical limit ≈ 20 kHz due to filter roll-off.
2. Resolution (Step Size ΔV): $V_{\text{FSR}} = 1 - (-1) = 2\,\text{V}$. $N = 16 \implies 2^N = 65536$ levels. $\Delta V = \frac{2\,\text{V}}{65536} \approx 30.5\,\mu\text{V}$.
3. SQNR (16-bit): SQNR ≈ $6.02N+1.76 = 6.02\times16+1.76 = 96.32+1.76 = 98.08$ dB. (Standard CD dynamic range ≈ 96 dB, close to this theoretical max).
4. SQNR (24-bit): $\text{SQNR}_{24} \approx 6.02 \times 24 + 1.76 = 144.48 + 1.76 = 146.24$ dB. Improvement = 146.24 - 98.08 = 48.16 dB. (Equivalent to 8 bits × 6.02 ≈ 48 dB).
5. Anti-Aliasing Filter: Why: Any frequency component $f > f_s/2$ in the input signal will be "mirrored" (aliased) into the baseband 0 to $f_s/2$ as a false frequency $f_{\text{alias}} = |f - kf_s|$. This corrupts the digital signal irreversibly. Characteristic: Low-pass filter. Cutoff $f_c \approx f_{\max}$ (e.g., 20 kHz). Stopband must start at $f_s - f_{\max} = 24.1$ kHz with high attenuation (> 96 dB for 16-bit) to prevent aliases folding into the passband. This requires a steep filter (high order) or oversampling techniques.

PROPERTY **Communication Channels: Bandwidth, Sidebands, Satellite, Base Stations**

- Channel Bandwidth: Range of frequencies allocated for a single communication channel. Must accommodate modulated signal BW (e.g., AM: 10 kHz, FM: 200 kHz, TV: 6 MHz/8 MHz, GSM: 200 kHz).
- Sidebands: Frequency components generated by modulation symmetrically around f_c.
  - AM: Two sidebands (USB, LSB) + Carrier (DSB-FC). BW = $2f_m$.
  - SSB (Single Sideband): Suppress carrier and one sideband. BW = f_m. Power efficient, complex receiver (coherent detection).
  - FM: Infinite sidebands theoretically; Carson's BW ($2(\Delta f + f_m)$) used practically.
- Satellite Communication (Geostationary - GEO):
  - Orbit: ≈ 35786 km above equator. Period = 24 h. Appears fixed.
  - Uplink (f_u) / Downlink (f_d): Frequencies differ (e.g., C-band: 6/4 GHz, Ku-band: 14/12 GHz) to avoid interference (Tx leakage into Rx).
  - Transponder Block Diagram: Rx Antenna → LNA (Low Noise Amp) → Bandpass Filter → Mixer + LO (Frequency Conversion $f_u \rightarrow f_d$) → HPA (High Power Amp, TWTA/SSPA) → Tx Antenna.

- Free Space Path Loss (FSPL):

$$L_p = \left(\frac{4\pi d}{\lambda}\right)^2 = \left(\frac{4\pi d f}{c}\right)^2$$

  In dB (d in km, f in GHz): $L_p(\text{dB}) = 20\log_{10}(d) + 20\log_{10}(f) + 92.45$.
- Link Budget: $P_R(\text{dBW}) = \text{EIRP} + G_R - L_p - L_{\text{atm}} - L_{\text{other}}$. EIRP $= P_T + G_T$ (dBW). G_R = Rx Antenna Gain (dBi).
- Latency: Round trip ≈ 240 ms (propagation only). Affects voice/TCP.

- Base Station (Cellular): Fixed transceiver serving a Cell. Functions: Radio Resource Management, Handover control, Connection to Mobile Switching Centre (MSC)/Core Network.
- Frequency Reuse: Same frequency channels reused in geographically separated cells (Co-channel Interference CCI managed). Cluster size K (e.g., 3, 4, 7, 12). Capacity ∝ Number of Clusters. Sectoring (120°/60°) reduces CCI → smaller K → higher capacity.

WORKED EXAMPLE 15 **Satellite Link Budget and Cellular Concepts**

A Geostationary satellite operates in Ku-band. Uplink f_u = 14 GHz, Downlink f_d = 12 GHz. Satellite distance d = 36000 km. Earth station antenna gain G_T = 50 dBi, Satellite receive antenna gain G_R = 30 dBi. Transmitter EIRP = 60 dBW.

1. Calculate the Free Space Path Loss (FSPL) for the Uplink in dB.
2. Calculate the Received Power at the Satellite (P_R) in dBW.
3. If the system uses Frequency Division Multiple Access (FDMA) with a total transponder bandwidth of 500 MHz and each channel requires 36 MHz (including guard bands), how many channels can be supported?
4. Explain the concept of "Frequency Reuse" in cellular networks and how it increases capacity.

SOLUTION

Detailed Solution

1. Uplink FSPL: Formula: $L_p(\text{dB}) = 20\log_{10}(d_{\text{km}}) + 20\log_{10}(f_{\text{GHz}}) + 92.45$. d = 36000 km, f = 14 GHz. $20\log_{10}(36000) = 20 \times 4.5563 = 91.126$. $20\log_{10}(14) = 20 \times 1.1461 = 22.922$. $L_p = 91.126 + 22.922 + 92.45 = 206.50$ dB.
2. Received Power at Satellite (P_R): Link Budget: $P_R(\text{dBW}) = \text{EIRP} + G_R - L_p$. EIRP = 60 dBW (Given as combined $P_T + G_T$). G_R = 30 dBi. $P_R = 60 + 30 - 206.5 = -116.5$ dBW. Convert

to Watts: $P_R = 10^{-11.65} \approx 2.24 \times 10^{-12}$ W = 2.24 pW. Note: This tiny power requires a very Low Noise Amplifier (LNA) at the satellite (Noise Figure ≈ 1 dB).

3. FDMA Channel Capacity: Total BW = 500 MHz. Channel BW = 36 MHz. $N = \left\lfloor \frac{500}{36} \right\rfloor = \lfloor 13.88 \rfloor =$ **13** channels. (Guard bands consume the remainder).

4. Frequency Reuse in Cellular Networks: Concept: The same set of frequency channels is reused in different cells that are geographically separated far enough so that co-channel interference (CCI) is below an acceptable threshold (Carrier-to-Interference ratio C/I). Mechanism: Service area divided into clusters of K cells (e.g., $K = 3, 4, 7, 12$). Each cell in a cluster gets a unique subset of the total channels. The cluster pattern repeats spatially. Capacity Increase: Total system capacity $C = M \times N_c$ where M = number of clusters, N_c = channels per cluster (N_{total}/K). Without reuse ($K = 1$), $C = N_{\text{total}}$. With reuse, $C = M \times (N_{\text{total}}/K)$. As area grows, M grows $\Longrightarrow$ Capacity grows linearly with area. This is the fundamental principle allowing millions of users with limited spectrum. Trade-off: Smaller K (tighter reuse) → Higher capacity but higher interference → Requires sectoring (directional antennas) / power control / advanced coding.

METHOD **Synthesis: Complete Link Analysis – Shannon-Hartley & Link Budget**

1. Shannon-Hartley Theorem: Maximum theoretical channel capacity (error-free) for a bandlimited AWGN channel:

$$C = B \log_2\left(1 + \frac{S}{N}\right) \quad \text{bits/s}$$

B = Bandwidth (Hz), S/N = Signal-to-Noise Ratio (linear power ratio).

2. Thermal Noise Power: $N = kTB$. $k = 1.38 \times 10^{-23}$ J/K (Boltzmann). T = Noise Temperature (K), typically 290 K (room temp). $N(\text{dBm}) = -174 + 10\log_{10}(B/\text{Hz}) + NF$ (Noise Figure in dB).

3. Energy per Bit: $E_b = S/R_b$. $N_0 = kT$. $\frac{E_b}{N_0} = \frac{S}{N}\frac{B}{R_b}$. Shannon Limit: $\frac{E_b}{N_0} > \ln 2 \approx -1.6$ dB for $R_b \to C$.

4. Link Budget Steps: 1. Tx Power (P_T) + Tx Antenna Gain (G_T) = EIRP (dBW/dBm). 2. -Path Loss (L_p) - Atmospheric/Misc Losses (L_a). 3. +Rx Antenna Gain (G_R). 4. = Received Power (P_R). 5. Noise Power $N = kT_{\text{sys}}B$ (dBm). 6. SNR = $P_R - N$ (in dB). 7. Check if SNR > SNR_{req} for modulation/coding (e.g., BPSK ≈ 10 dB E_b/N_0 for BER 10^{-6}).

5. Digital Modulation Spectral Efficiency: $\eta = R_b/B$ (bits/s/Hz). BPSK=1, QPSK=2, 16-QAM=4, 64-QAM=6. Higher η requires higher SNR.

WORKED EXAMPLE 16 **Shannon Capacity and Link Budget for a Digital Radio Link**

A point-to-point microwave link operates at 10 GHz over 30 km. Bandwidth B = 20 MHz. Tx Power P_T = 1 W (30 dBm). Tx/Rx Antenna Gains = 35 dBi each. Receiver Noise Figure NF = 3 dB. System uses QPSK (R_b = 40 Mbps). Required E_b/N_0 = 10 dB for target BER.

1. Calculate Free Space Path Loss (FSPL).
2. Calculate Received Power P_R (dBm).
3. Calculate Receiver Noise Power N (dBm).
4. Determine achieved E_b/N_0 (dB). Is the link viable? What is the Fade Margin?
5. What is the Shannon Capacity of this channel? Is QPSK spectral efficiency close to the limit?

SOLUTION

Detailed Solution

1. FSPL: $L_p = 20\log_{10}(30) + 20\log_{10}(10) + 92.45 = 29.54 + 20 + 92.45 = 141.99$ dB ≈ 142 dB.
2. Received Power P_R: EIRP $= P_T + G_T = 30 + 35 = 65$ dBm. $P_R = \text{EIRP} - L_p + G_R = 65 - 142 + 35 = -42$ dBm.
3. Noise Power N: $N(\text{dBm}) = -174 + 10\log_{10}(B) + NF$. $10\log_{10}(20 \times 10^6) = 10 \times 7.3010 = 73.01$ dB. $N = -174 + 73.01 + 3 = -97.99$ dBm ≈ -98 dBm.
4. Achieved E_b/N_0 and Viability: SNR $= P_R - N = -42 - (-98) = 56$ dB. R_b = 40 Mbps $\Longrightarrow$ $10\log_{10}(R_b) = 76.02$ dB. $10\log_{10}(B) = 73.01$ dB. $\frac{E_b}{N_0}(\text{dB}) = \text{SNR} + 10\log_{10}(B/R_b) = 56 + (73.01 - 76.02) = 56 - 3.01 = 52.99$ dB. Required E_b/N_0 = 10 dB. Fade Margin = Achieved - Required = 53 - 10 = 43 dB. Conclusion: Link is highly viable with enormous fade margin (43 dB). Typical design targets 20 dB–30 dB margin.
5. Shannon Capacity: $\text{SNR}_{\text{lin}} = 10^{56/10} \approx 3.98 \times 10^5$. $C = B\log_2(1+\text{SNR}) = 20 \times 10^6 \times \log_2(3.98 \times 10^5)$. $\log_2(3.98 \times 10^5) = \frac{\log_{10}(3.98 \times 10^5)}{\log_{10} 2} \approx \frac{5.600}{0.3010} \approx 18.60$. $C \approx 20 \times 10^6 \times 18.60 = 372$ Mbps. Spectral Efficiency: Shannon Limit $\eta_{\max} = C/B \approx 18.6$ bit/s/Hz. QPSK Efficiency $\eta_{\text{QPSK}} = 2$ bit/s/Hz. Analysis: QPSK uses only $\approx 10\%$ of the theoretical capacity. Higher order modulation (e.g., 256–QAM, $\eta = 8$) or coding could increase throughput significantly, but would require higher SNR (reducing fade margin). The current link is "power limited" (excess power) but "bandwidth limited" (fixed 20 MHz).

DID YOU KNOW? **Cameroon Context: Communication Infrastructure**

- Satellite: Cameroon uses Intelsat, Eutelsat, RASCOM (African satellite) for international

trunking, TV distribution (CRTV), and VSAT networks in remote areas (e.g., Far North, East regions).

- Fibre Optic: CAMTEL manages the national backbone (WACS, SAT-3 submarine cable landing points in Kribi/Limbe). Major cities (Yaoundé, Douala, Bafoussam, Garoua) linked by fibre.
- Mobile: MTN Cameroon, Orange Cameroon, Nexttel (Viettel), Camtel Mobile. Technologies: GSM (900/1800 MHz), UMTS (2100 MHz), LTE (800/1800/2600 MHz). Base stations (BTS/NodeB/eNodeB) ubiquitous.
- Regulation: ART (Agence de Régulation des Télécommunications) manages spectrum allocation, licensing, numbering.
- Broadcast: CRTV (AM/MF: 918 kHz Yaoundé; FM/VHF: 88–108 MHz nationwide; TV/UHF). Private radio/TV stations licensed by ART.

WARNING **Common Examination Traps**

- AM Modulation Index: $m = E_m / E_c$. Never exceed 1. If V_{max}, V_{min} given on CRO: $m = (V_{max} - V_{min})/(V_{max} + V_{min})$.
- FM Modulation Index: $\beta = \Delta f / f_m$. Can be > 1. Do not confuse with AM m.
- Bandwidth: AM = $2f_m$. FM = $2(\Delta f + f_m)$ (Carson). Don't use $2f_m$ for FM.
- Tuning Circuit Q: $Q = \omega L / r$ (series r). If parallel R_p given, $Q = R_p/(\omega L)$. Check circuit diagram!
- Envelope Detector RC: $\frac{1}{\omega_c} \ll RC \ll \frac{1}{\omega_m}$. Geometric mean $\sqrt{\frac{1}{\omega_c \omega_m}}$ is a good design choice.
- Satellite Link Budget: Uplink and Downlink frequencies differ → Path Loss differs. Use correct f for each leg.
- Shannon Capacity: $C = B \log_2(1 + S/N)$. S/N is linear power ratio, not dB. Convert dB → linear first.
- Units: Always write units (Hz, W, dB, dBm, dBW, Ω, F, H). dBm ref 1 mW, dBW ref 1 W.

8 EXERCISES

8.1 I check my knowledge

EXERCISE 5 **MCQ: Radio wave propagation**

Which of the following statements about radio wave propagation is correct?

1. Ground waves follow the curvature of the Earth and are most effective at low frequencies.
2. Sky waves are reflected by the ionosphere and are used for very high frequency (VHF) communication.
3. Space waves travel in straight lines and are the only mode used for satellite communication.
4. The critical frequency is the highest frequency that can be reflected by the ionosphere at vertical incidence.

EXERCISE 6 **True/False with justification**

State whether each statement is true or false and justify your answer briefly.

1. In amplitude modulation (AM), the bandwidth required is twice the maximum modulating frequency.
2. Frequency modulation (FM) is more susceptible to noise than AM because noise affects frequency variations.
3. A half-wave dipole aerial has a length equal to one wavelength of the transmitted signal.
4. The Nyquist sampling theorem states that the sampling frequency must be at least equal to the highest frequency present in the signal.

EXERCISE 7 **Definitions**

Define the following terms as used in communication systems:

1. Modulation index (for AM)
2. Sidebands

3. Quantisation error
4. Geostationary satellite

EXERCISE 8 **Direct calculation: Wavelength and aerial length**

A radio station broadcasts on a carrier frequency of 95.5 MHz. Calculate:

1. The wavelength of the carrier wave in metres.
2. The length of a half-wave dipole aerial for this frequency.

8.2 I practise

EXERCISE 9 **AM modulation index and overmodulation**

An AM wave displayed on an oscilloscope has a maximum envelope amplitude of 12 V and a minimum envelope amplitude of 4 V.

1. Calculate the modulation index m.
2. State whether the signal is overmodulated. Justify your answer.
3. If the carrier amplitude is 8 V, determine the peak amplitude of each sideband.

EXERCISE 10 **Tuning circuit frequency range**

A radio receiver uses a tuning circuit consisting of a fixed inductor of inductance L = 180 μH and a variable capacitor adjustable from C_{min} = 20 pF to C_{max} = 300 pF.

1. Determine the minimum and maximum resonant frequencies of the circuit.
2. Identify the radio band (LW, MW, SW, VHF/FM) covered by this range.
3. Sketch the resonance curve (current vs frequency) for a given capacitance, labelling the resonant frequency, bandwidth and quality factor Q.

EXERCISE 11 **AM frequency spectrum and bandwidth**

An audio signal containing frequencies up to 4.5 kHz amplitude-modulates a carrier of frequency 1 MHz.

1. Sketch the frequency spectrum of the resulting AM wave, clearly marking the carrier and the two sidebands.
2. Calculate the channel bandwidth required.
3. If the same audio signal were used for FM with a maximum frequency deviation of 75 kHz, estimate the bandwidth using Carson's rule.

EXERCISE 12 **Digital audio parameters**

A music signal is sampled at 44.1 kHz with 16-bit quantisation.

1. Calculate the number of quantisation levels.
2. Determine the bit rate (in kbps) for a stereo signal (two channels).
3. What is the maximum audio frequency that can be faithfully reproduced according to the Nyquist criterion?
4. Explain why a low-pass anti-aliasing filter is needed before sampling.

8.3 I prepare for the GCE

EXERCISE 13 **Propagation modes and reception behind a hill] [15 marks**

A car radio receives both MW (medium wave, 1 MHz) and VHF/FM (100 MHz) broadcasts. When the car drives behind a steep hill, the FM signal is lost while the MW signal remains audible.

1. Name and describe the three main modes of radio wave propagation. [6 marks]
2. Using diagrams, explain why the MW signal can still be received behind the hill while the FM signal cannot. [6 marks]
3. State one advantage and one disadvantage of using sky wave propagation for long-distance communication. [3 marks]

EXERCISE 14 **Comparison of AM and FM; TV sound transmission] [15 marks**

1. Compare amplitude modulation (AM) and frequency modulation (FM) under the following four headings:
   - Susceptibility to noise

- Bandwidth requirement
- Range (for a given transmitter power)
- Complexity of transmitter and receiver circuits

[8 marks]

2. The Cameroon Radio Television (CRTV) transmits television sound using FM rather than AM. Justify this choice with reference to the comparison above. [4 marks]
3. An FM transmitter has a carrier frequency of 98.5 MHz and a maximum frequency deviation of 75 kHz. If the modulating signal has a maximum frequency of 15 kHz, calculate the approximate bandwidth using Carson's rule. [3 marks]

EXERCISE 15 **Satellite communication delay] [12 marks**

A geostationary satellite orbits at an altitude of 36,000 km above the Earth's surface. A telephone call is routed from Douala (Cameroon) to London (UK) via this satellite. Assume the uplink and downlink paths are each approximately equal to the satellite altitude (ignore Earth curvature for this calculation).

1. Calculate the minimum time delay for a signal to travel from Douala to the satellite and then to London (one-way). [4 marks]
2. If the conversation is a two-way telephone call, what is the total round-trip delay experienced by the speakers? [2 marks]
3. Comment on the effect of this delay on natural telephone conversation. [3 marks]
4. Suggest one technical method to mitigate the annoyance caused by this delay in satellite telephony. [3 marks]

9 SOLUTIONS TO THE EXERCISES

10 CORRECTED EXERCISES — COMMUNICATION (PHY14)

SOLUTION **Exercise 1 — MCQ: Radio wave propagation**

Correct answers: (a) and (d).

- (a) Correct. Ground (surface) waves travel along the Earth's surface, following its curvature by diffraction. Diffraction is significant only when the wavelength is comparable to obstacles, i.e. at **low frequencies** (long wavelengths, LW and MW bands).
- (b) False. Sky waves are indeed reflected (refracted) back by the ionosphere, but this works for HF (short wave, roughly 3–30 MHz). VHF waves (30–300 MHz) pass through the ionosphere and are not reflected; VHF uses the space wave.
- (c) False. Space waves do travel in straight lines (line of sight) and are used for satellite links, but they are not the only mode used for satellites in general; the word "only" makes the statement incorrect.
- (d) Correct. The **critical frequency** is by definition the highest frequency reflected back to Earth when the wave is sent vertically (at vertical incidence) towards the ionosphere. Above it, the wave penetrates the layer and escapes into space.

SOLUTION **Exercise 2 — True/False with justification**

1. True. An AM wave contains the carrier at f_c and two sidebands extending from $f_c - f_m$ to $f_c + f_m$, where f_m is the maximum modulating frequency. Hence the bandwidth is
$$B = (f_c + f_m) - (f_c - f_m) = 2f_m.$$
2. False. It is the opposite: FM is less susceptible to noise. Noise and interference mainly add unwanted amplitude variations; in FM the information is carried by frequency variations, and amplitude limiters in the receiver remove the noise. In AM, noise directly corrupts the amplitude that carries the information.
3. False. A half-wave dipole has a total length $L = \lambda/2$, i.e. half a wavelength, not a full wavelength.
4. False. The Nyquist theorem requires the sampling frequency to be at least twice the highest frequency present: $f_s \geq 2f_{\max}$. Sampling at exactly $f_{\max}$ would cause severe loss of information (aliasing).

SOLUTION **Exercise 3 — Definitions**

1. Modulation index (AM): the ratio of the amplitude of the modulating signal to the amplitude of the carrier,
$$m = \frac{A_m}{A_c},$$
equivalently $m = \frac{A_{max} - A_{min}}{A_{max} + A_{min}}$ from the envelope. It is dimensionless; $m \leq 1$ for distortion-free AM.
2. Sidebands: the bands of new frequencies produced by modulation, symmetrically placed on either side of the carrier: the upper sideband (USB) around $f_c + f_m$ and the lower sideband (LSB) around $f_c - f_m$. They carry all the information of the modulating signal.
3. Quantisation error: the difference between the actual value of the analogue sample and the nearest quantisation level (digital value) assigned to it. It is at most half a quantisation step, $\pm q/2$, and appears as quantisation noise.
4. Geostationary satellite: a satellite in a circular equatorial orbit at an altitude of about 36 000 km, with an orbital period equal to the Earth's rotation period (24 h). It therefore appears fixed above a point on the equator, allowing fixed ground aerials to communicate with it continuously.

SOLUTION **Exercise 4 — Direct calculation: Wavelength and aerial length**

Given: $f = 95.5\ \text{MHz} = 9.55 \times 10^7\ \text{Hz}$; $c = 3.00 \times 10^8\ \text{m s}^{-1}$.

1. Wavelength:
$$\lambda = \frac{c}{f} = \frac{3.00 \times 10^8}{9.55 \times 10^7} = 3.14\ \text{m}.$$
2. A half-wave dipole has length $L = \lambda/2$:
$$L = \frac{3.14}{2} = 1.57\ \text{m}.$$

This is why FM radio aerials (telescopic rods on radios, or rooftop aerials) are of the order of a metre long.

SOLUTION **Exercise 5 — AM modulation index and overmodulation**

Given: $A_{max} = 12\ \text{V}$, $A_{min} = 4\ \text{V}$.

1. The modulation index is obtained from the envelope:

$$m = \frac{A_{\max} - A_{\min}}{A_{\max} + A_{\min}} = \frac{12 - 4}{12 + 4} = \frac{8}{16} = 0.50.$$

2. The signal is not overmodulated, since $m = 0.50 < 1$. Overmodulation occurs when $m > 1$; the envelope then crosses zero and the message is distorted on demodulation.

3. The carrier amplitude is $A_c = \frac{A_{\max} + A_{\min}}{2} = 8\,\text{V}$ (consistent with the data). Each sideband has peak amplitude

$$A_{\text{SB}} = \frac{mA_c}{2} = \frac{0.50 \times 8}{2} = 2.0\,\text{V}.$$

WARNING **Common mistake**

Do not write $A_{\text{SB}} = mA_c$: the modulating signal's energy is shared equally between the two sidebands, so each gets $mA_c/2$.

SOLUTION **Exercise 6 — Tuning circuit frequency range**

The resonant frequency of an LC circuit is

$$f_0 = \frac{1}{2\pi\sqrt{LC}}.$$

Since $f_0 \propto 1/\sqrt{C}$, the maximum frequency corresponds to $C_{\min}$ and vice versa.

1. Maximum frequency ($C_{\min} = 20$ pF):

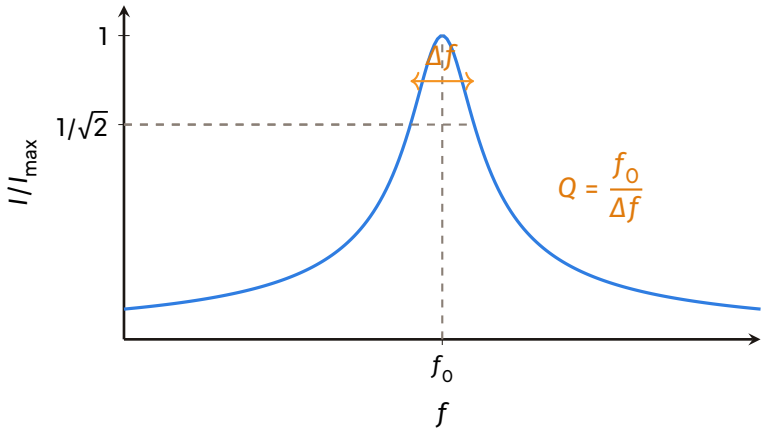
$$f_{\max} = \frac{1}{2\pi\sqrt{180 \times 10^{-6} \times 20 \times 10^{-12}}} = \frac{1}{2\pi\sqrt{3.6 \times 10^{-15}}} \approx 2.65\,\text{MHz}.$$

   Minimum frequency ($C_{\max} = 300$ pF):

$$f_{\min} = \frac{1}{2\pi\sqrt{180 \times 10^{-6} \times 300 \times 10^{-12}}} = \frac{1}{2\pi\sqrt{5.4 \times 10^{-14}}} \approx 0.685\,\text{MHz} = 685\,\text{kHz}.$$

2. The range 0.685–2.65 MHz covers the medium wave (MW) band (roughly 0.53–1.6 MHz) and extends into the lower short wave (SW) band. It does not reach LW (< 300 kHz) or VHF/FM (88–108 MHz).

3. Resonance curve: the current I is maximum at f_0; the bandwidth Δf is measured between the half-power points ($I = I_{\max}/\sqrt{2}$), and $Q = f_0/\Delta f$.

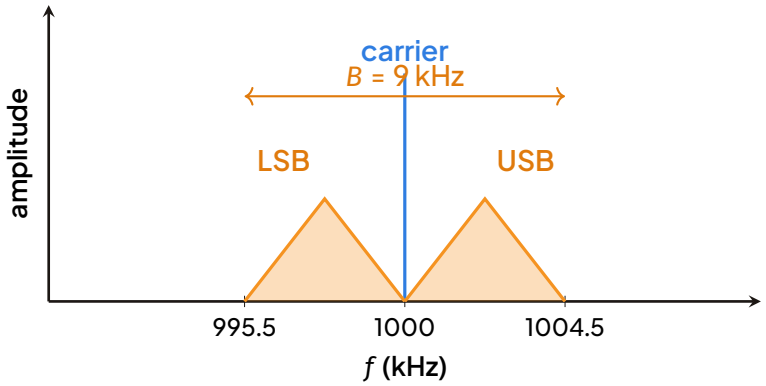


Resonance curve of the tuning circuit: sharpness set by the quality factor Q. Half-power points at $f_0 \pm \Delta f/2$.

SOLUTION **Exercise 7 — AM frequency spectrum and bandwidth**

Given: f_c = 1 MHz = 1000 kHz, f_m(max) = 4.5 kHz.

1. The spectrum contains the carrier at 1000 kHz, a lower sideband from 1000 − 4.5 = 995.5 kHz up to the carrier, and an upper sideband from the carrier to 1000 + 4.5 = 1004.5 kHz.



Frequency spectrum of the AM wave: carrier and two sidebands.

2. Channel bandwidth:

$$B_{AM} = 2f_m = 2 \times 4.5 = 9.0 \text{ kHz}.$$

3. Carson's rule for FM (assuming standard broadcast deviation Δf = 75 kHz):

$$B_{FM} \approx 2(\Delta f + f_m) = 2(75 + 4.5) = 159 \text{ kHz}.$$

Comment: FM needs nearly 18 times more bandwidth than AM for the same audio signal — the price paid for its superior noise immunity.

SOLUTION **Exercise 8 — Digital audio parameters**

Given: f_s = 44.1 kHz, n = 16 bits, 2 channels (stereo).

1. Number of quantisation levels:

$$N = 2^n = 2^{16} = 65\,536 \text{ levels}.$$

2. Bit rate for stereo:

$$R = n \times f_s \times 2 = 16 \times 44.1 \times 10^3 \times 2 = 1.4112 \times 10^6 \text{ bit s}^{-1} = 1411 \text{ kbps}.$$

(This is exactly the bit rate of audio CD.)

3. Nyquist criterion: $f_{max} = f_s/2 = 44.1/2 = 22.05$ kHz. This comfortably covers the human hearing range (up to about 20 kHz).

4. Any frequency component above $f_s/2$ present at the sampler input is "folded back" into the audio band as a false low frequency (**aliasing**); once sampled, this corruption cannot be removed. A low-pass anti-aliasing filter with cut-off at about 22 kHz is therefore placed before the ADC to eliminate all components that could alias, guaranteeing faithful reconstruction.

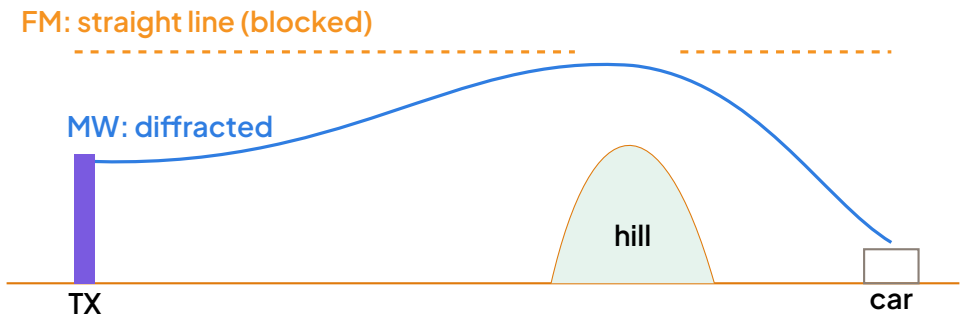
SOLUTION **Exercise 9 — Propagation modes and reception behind a hill (15 marks)**

(a) The three propagation modes [6 marks — 2 per mode: name 1, description 1]

- Ground (surface) wave: the wave travels along the Earth's surface, following its curvature by diffraction around obstacles. Effective at low frequencies (LW, MW); range limited to a few hundred km by ground absorption.
- Sky wave: the wave is directed upwards and refracted/reflected back to Earth by the ionosphere. Used in HF (short wave, 3–30 MHz); enables intercontinental communication by successive hops, but is variable with time of day and solar activity.
- Space (line-of-sight) wave: the wave travels in a straight line from transmitting to receiving aerial, including reflection from the ground. Used at VHF/UHF and above (FM

radio, TV, satellite links); range limited by the horizon and obstacles.

(b) Why MW survives behind the hill but FM does not [6 marks]
MW (~ 1 MHz) has $\lambda = c/f \approx 300$ m, comparable to the size of the hill: the wave diffracts strongly around and over the hill and also propagates as a ground wave hugging the terrain, so it reaches the car.
VHF/FM (~ 100 MHz) has $\lambda \approx 3$ m, far smaller than the hill: diffraction is negligible, the wave travels essentially in a straight line, and the hill casts a radio shadow in which no signal is received.



Long-wave MW signal diffracts over the hill; the short-wavelength FM signal is blocked.

Examiner expectations: a labelled diagram, the comparison of λ with obstacle size, and the key word diffraction.
(c) Sky wave: one advantage, one disadvantage [3 marks]

- Advantage: very long-distance (intercontinental) communication possible without satellites or repeaters, using ionospheric reflection.
- Disadvantage: unreliable and variable — the ionosphere changes with time of day, season and solar activity, causing fading and sometimes total loss of signal.

SOLUTION **Exercise 10 — Comparison of AM and FM; TV sound transmission (15 marks)**

(a) Comparison [8 marks — 2 per heading]

Criterion	AM	FM
Noise susceptibility	High: noise adds amplitude variations which directly corrupt the message	Low: information is in frequency; amplitude limiters remove noise
Bandwidth	Narrow: $B = 2f_m$ (e.g. 9 kHz)	Wide: $B \approx 2(\Delta f + f_m)$ (e.g. 180 kHz)
Range (same power)	Greater: ground/sky waves at LW–MW–HF cover long distances	Smaller: VHF space wave limited roughly to line of sight
Circuit complexity	Simple, cheap transmitter and receiver (diode detector)	More complex, costlier (limiter, discriminator/PLL)

(b) Why CRTV uses FM for TV sound [4 marks]

- Television sound must be of high fidelity; FM's superior noise immunity gives clear, hiss-free audio, whereas AM would crackle with every interference (motors, lightning, electrical appliances).
- TV already operates at VHF/UHF where wide channels are available, so FM's large bandwidth is affordable.
- The wide FM bandwidth accommodates the full audio range (15 kHz) with a large deviation, giving a high signal-to-noise ratio.
- AM's only real advantages (long range, simplicity) are irrelevant here since TV is broadcast from local transmitters within line of sight.

(c) Carson's rule [3 marks]

$$B \approx 2(\Delta f + f_m) = 2(75 + 15) = 2 \times 90 = 180\ \text{kHz}.$$

Mark scheme: correct formula [1], correct substitution [1], answer 180 kHz with unit [1].

SOLUTION **Exercise 11 — Satellite communication delay (12 marks)**

Given: altitude $h = 36\,000$ km $= 3.6 \times 10^7$ m; $c = 3.00 \times 10^8$ m s^{-1}.
(a) One-way delay [4 marks]
The path is Douala → satellite → London, i.e. two hops each of length approximately h (standard approximation for geostationary links):

$$d = 2h = 7.2 \times 10^7\ \text{m}.$$

$$t = \frac{d}{c} = \frac{7.2 \times 10^7}{3.00 \times 10^8} = 0.24\ \text{s}.$$

Mark scheme: identifying two hops [1], $d = 2h$ [1], formula $t = d/c$ [1], answer 0.24 s [1].

(b) Round-trip delay [2 marks]
A reply must make the same journey back:

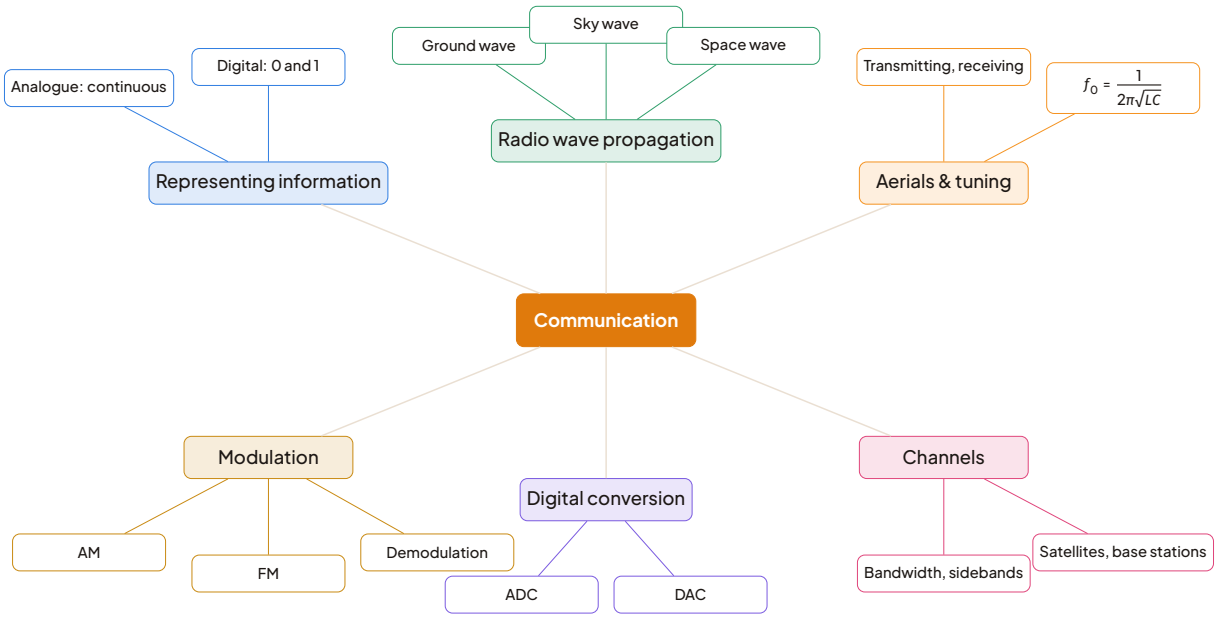
$$t_{RT} = 2 \times 0.24 = 0.48\ \text{s} \approx 0.5\ \text{s}.$$

(c) Effect on conversation [3 marks]
A pause of about half a second between speaking and hearing the reply is clearly perceptible. Speakers begin to talk over each other, hesitate, or repeat themselves, because the natural rhythm of conversation expects replies within a small fraction of a second. The conversation feels unnatural and tiring.
(d) Mitigation [3 marks]
Any one of:

- Echo cancellers and digital signal processing at the exchanges, so that the delayed echo of one's own voice (the most disturbing effect) is suppressed.
- Routing the call via undersea fibre-optic cables (or low-Earth-orbit satellite constellations) instead of a geostationary satellite, drastically shortening the path and the delay.
- Disciplined "push-to-talk" style protocols where each speaker waits for a clear end of transmission.

Examiner expectation: one well-explained technical method suffices for full marks.

11 SUMMARY SHEET



Concept map of the chapter Communication: the six themes and their key ideas.

KEY POINTS **Communication — Summary Sheet**

1. Key definitions

- **Analogue signal**: varies continuously with time, taking any value within a range (e.g. the voltage from a microphone).
- **Digital signal**: takes only discrete values, usually two levels (0 and 1); information is coded in binary.
- **Carrier wave**: high-frequency radio wave that "carries" the low-frequency information signal.
- **Modulation**: process of superimposing the information signal onto the carrier by varying one of its characteristics (amplitude or frequency).
- **Demodulation** (detection): recovering the original information signal from the modulated carrier at the receiver.
- **Bandwidth**: range of frequencies occupied by a signal or passed by a channel, $\Delta f = f_{max} - f_{min}$.
- **Sidebands**: bands of frequencies above and below the carrier frequency f_c, produced by modulation.

- **Base station**: fixed transmitter–receiver linking mobile phones to the telephone network within its cell.

2. Laws and formulas to know

- Wave equation: $c = f\lambda$ with $c = 3.00e8\ \mathrm{m\,s^{-1}}$ for all electromagnetic waves in air or vacuum.
- Resonance of the LC tuning circuit: $f_0 = \frac{1}{2\pi\sqrt{LC}}$; at resonance the response is maximum (peak of the resonance curve). The sharper the peak, the better the **selectivity**.
- AM: carrier amplitude varies with the signal; modulation index $m = \frac{A_{\text{signal}}}{A_{\text{carrier}}}$; keep $m \leq 1$ to avoid distortion. Bandwidth = $2f_m$ (two sidebands $f_c \pm f_m$).
- FM: carrier frequency varies with the signal; amplitude constant; bandwidth larger than AM.
- Sampling: to digitise correctly, sample at $f_s \geq 2f_{\max}$ (sampling criterion).

3. Propagation of radio waves

- **Ground (surface) wave**: follows the Earth's surface; low frequencies; long range over land/sea (LW, MW).
- **Sky wave**: reflected by the ionosphere back to Earth; short waves; very long range, but varies with time of day.
- **Space wave**: straight line-of-sight between aerials; VHF/UHF (FM radio, TV, satellites); limited by the horizon, needs relay masts or satellites.

4. AM versus FM; analogue versus digital

- AM: simpler, cheaper, longer range, smaller bandwidth; but noisy and lower sound quality.
- FM: noise-resistant, better quality; but shorter range (space wave), wider bandwidth, costlier.
- Digital advantages: immune to noise, easy regeneration, storage and encryption; analogue is simpler and needs no conversion.

5. Methods

- To find the station received by a tuning circuit: compute $f_0 = \frac{1}{2\pi\sqrt{LC}}$ and compare with the carrier frequencies.
- To sketch AM: draw the carrier, then vary its amplitude following the signal envelope; for FM keep amplitude constant and vary the spacing of the oscillations.
- ADC chain: sampling → quantisation → binary coding; DAC reverses it.

6. Common mistakes

- Confusing carrier frequency f_c with signal frequency f_m; the sidebands are at $f_c \pm f_m$, not at f_m.
- Forgetting the 2π in $f_0 = \dfrac{1}{2\pi\sqrt{LC}}$.
- Saying FM changes the amplitude — it changes the frequency.
- Mixing up sky wave (ionosphere) and space wave (line of sight).

7. GCE tips

- Always give units (Hz, m) and use $c = 3.00e8\ \text{m s}^{-1}$; quote 3 significant figures.
- In structured questions, define terms before comparing (e.g. state what modulation is before contrasting AM and FM).
- Be ready to draw and label a resonance curve and an AM waveform — frequent exam items.

Signed • OnBuch+
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